



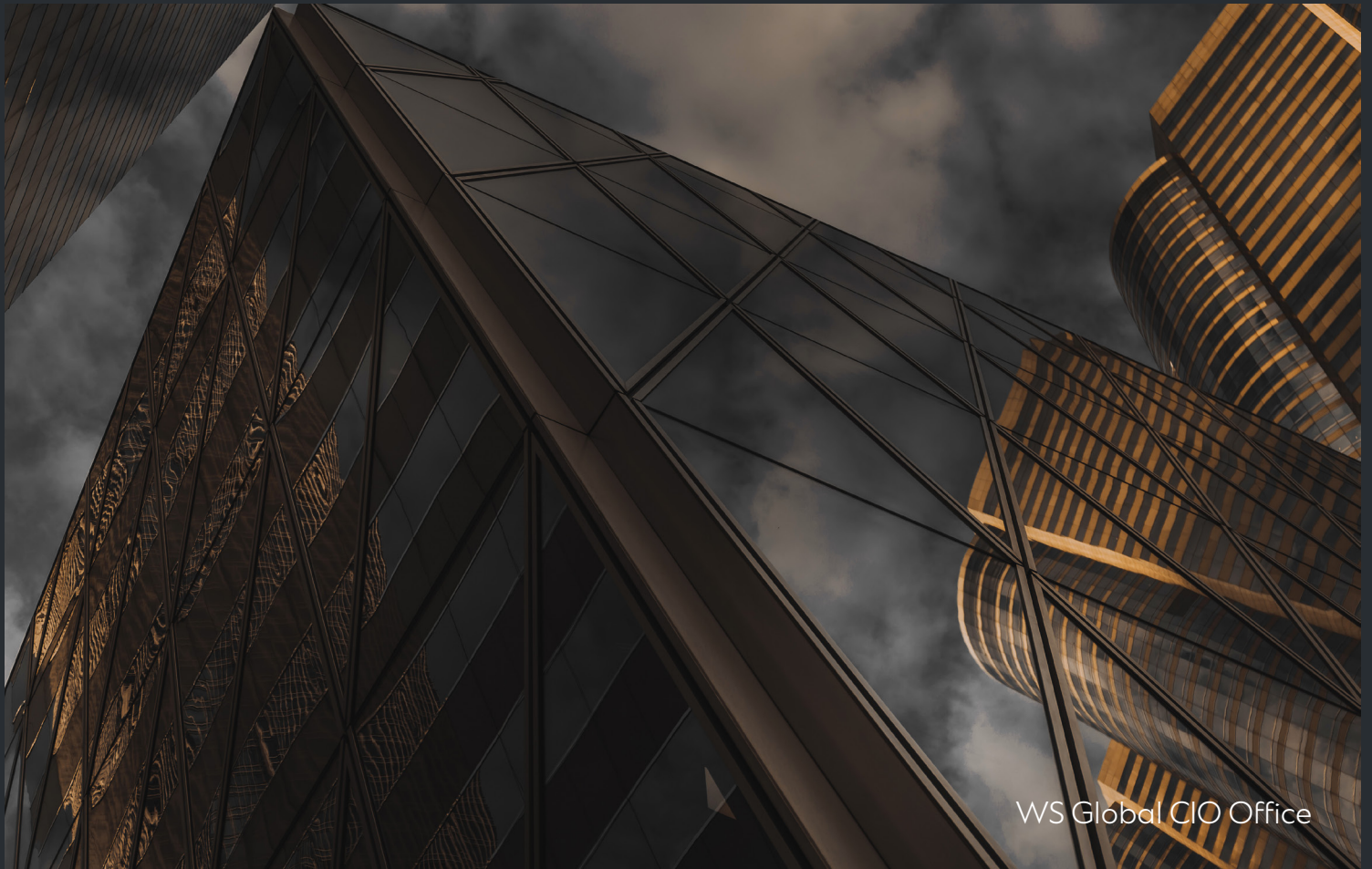
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Gold market dynamics

Revisiting the safe haven

April 2026



WS Global CIO Office

Gold's relevance in modern portfolios extends beyond its historical status as a store of value.

In an environment marked by recurring geopolitical frictions, macroeconomic fragmentation and shifting policy regimes, gold has demonstrated an enduring capacity to act as a long term portfolio stabiliser, lending resilience when other assets face synchronised drawdowns. This time-tested attribute continues to underpin its relevance in diversified investment strategies.

In this report, we explore gold's role as a hedge against market volatility and economic instability. We also synthesise insights from historical analyses and review gold's performance across market cycles. Furthermore, we discuss the various methods to include gold in a portfolio.

Overview of gold

The World Gold Council (WGC) estimated that the stock of gold above ground was approximately 220,000 tonnes as of 2025. Based on March-end 2026 gold prices, the stock's value stood at around USD 33trn, nearly 53% higher than a year ago, illustrative of the sharp rise in gold prices recently.

From an end-user perspective, gold remains a cornerstone in making jewellery and ornaments and showcasing wealth, which continue to be key demand drivers for this asset class. Historically, gold has been viewed as a currency, largely due to its role as a store of value. Gold is also increasingly incorporated in electrical equipment for longevity and improved performance. Importantly, however, gold has found its way into investment portfolios given its perceived quality of being an inflation hedge over long periods and its low correlation with equity and bonds.



Gold

- Known for stability, reliability
- Acts as an inflation hedge
- Holds low correlation with stocks and bonds (long term)
- Seen as a safe haven during a crisis

Source: Standard Chartered

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Why consider gold?

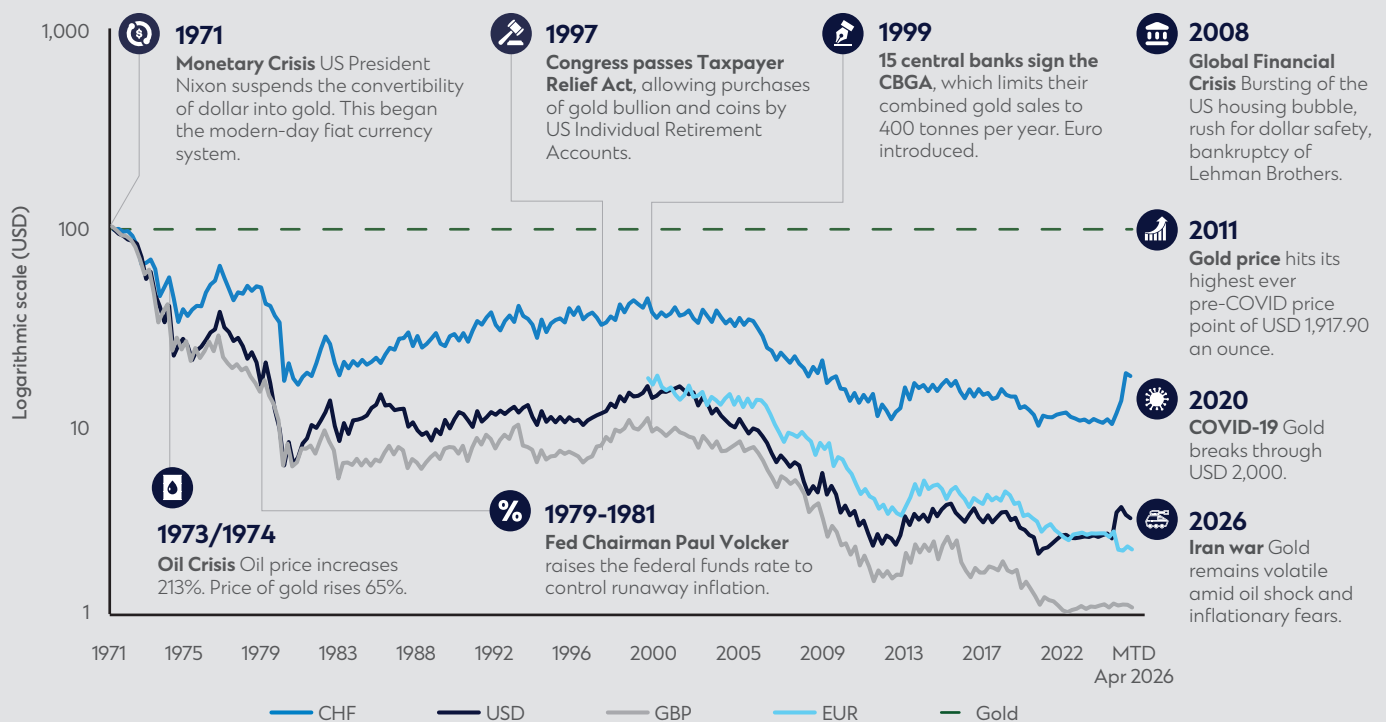
In this section, we examine how good gold is as an inflation hedge and a portfolio diversifier, and whether it deserves its safe haven status.

Inflation hedge

WGC data suggest that, over long periods (40+ years), gold can be a good inflation hedge. During inflationary periods, the purchasing power of fiat currency declines, but gold tends to hold, or grow, in value. Gold's naturally limited supply is an additional support. The inverse relationship between gold and major fiat currencies has held since the 1970s.

Fig. 1

Major fiat currencies vs gold



Source: Standard Chartered

One of the most inflationary periods in recent history was between 1973 and 1979, when average annual inflation in the US was around 8.2%. This was, of course, led by a jump in oil prices. During this period, gold gained almost 35% annually, an outcome supporting its status as a hedge against inflation. A recent study by WGC, analysing data from 1971 to 2020, found that gold delivered an annual growth rate of 8% to 10%, significantly outpacing CPI inflation of approximately 3.9%. Gold's value as a financial asset ties it more closely to economic growth and monetary expansion than to CPI alone, reinforcing its role as a long-term inflation hedge with positive real returns.

Note that while gold's performance was strong when inflation was high, some of this performance subsequently reversed once inflation was tamed via sharply higher interest rates as a result of tightening Fed policy. Once interest rates were raised sharply to bring inflation under control, gold failed as an effective hedge against inflation. For instance, between 1975 and 1985, a period of tightening US monetary policy stance, gold experienced five instances of negative annualised returns, with the largest such decline exceeding 30% in USD terms in 1981.

A more recent inflationary period was in 2021-22, when prices shot up as a result of COVID-led supply chain disruption, a jump in government spending and relatively low interest rates. US inflation hit a high of over 9% in June 2022. Gold hit a cyclical high of around USD 2,050 in March 2022, supporting the view that it acts as a hedge against inflation. However, the reverse also held – gold subsequently fell over 20% and did not bottom until October 2022. This fall in gold prices coincided with the Fed's aggressive interest rate hiking cycles. Higher bond yields (especially net of inflation) and a stronger USD are two factors that worked against gold – both in this instance and over longer periods of time. What this shows is that while gold remains an attractive hedge against rising inflation, tightening Fed policy aimed at taming inflation can hurt gold prices.

A similar paradox emerged in early 2026 during the Iran war, when gold hit an all time high of USD 5,589 in January 2026, before crashing nearly 25% by March 2026 to approximately USD 4,130, and finally rebounding to its long term uptrend. Despite geopolitical tensions that had driven safe-haven demand, oil shock, inflation fears and expectations of higher rates pressured gold downward.

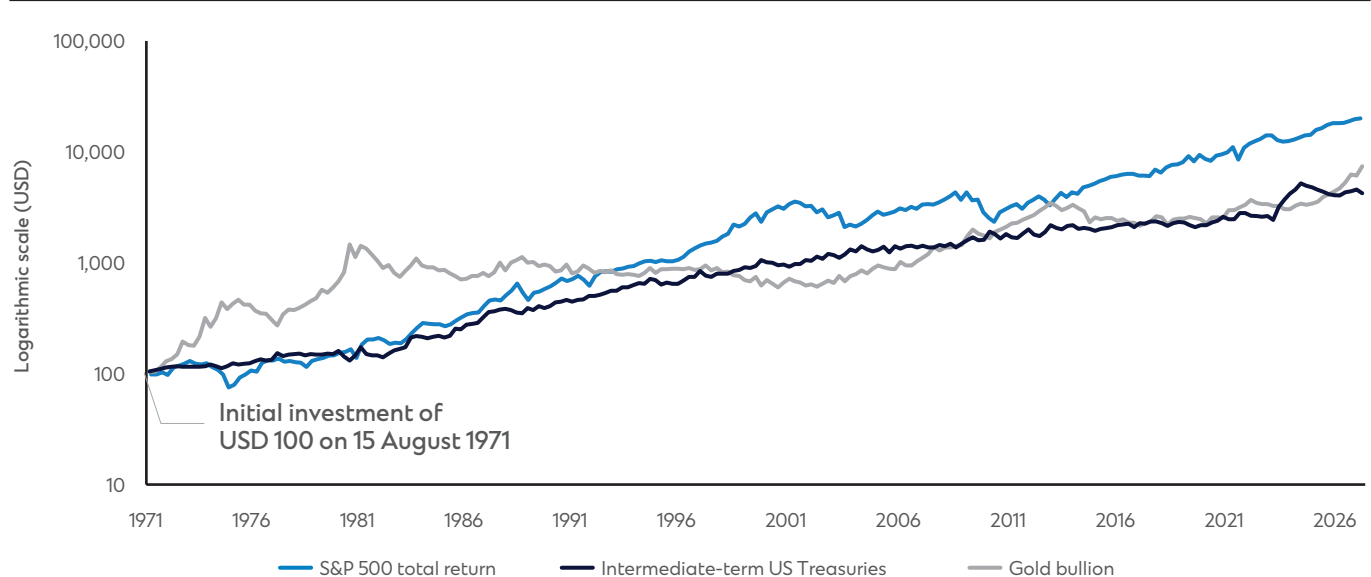
The aforesaid instances of gold's rise and subsequent fall indicate that gold may not be the 'perfect' safe haven at all times and that it too can succumb to the vagaries of economic and market cycles. However, over the past 50 years, the average annual rate of return from gold has been approximately 8%. Such a performance suggests that while gold can face temporary headwinds, it can potentially provide inflation-beating returns in the long term.

Portfolio diversification

Over the past 50+ years, gold has shared a low correlation with equities. Meanwhile, gold's correlation with US government bonds is generally negative during periods of high inflation, as bond prices tend to fall but gold prices rise during such periods. During crises, however, markets instead view both gold and US government bonds as safe haven assets, resulting in a slightly positive correlation between them. Gold's overall correlation with bonds in the long term, however, is low.

Fig. 2

Gold, bonds and equities



Source: TopForeignStocks.com, Standard Chartered

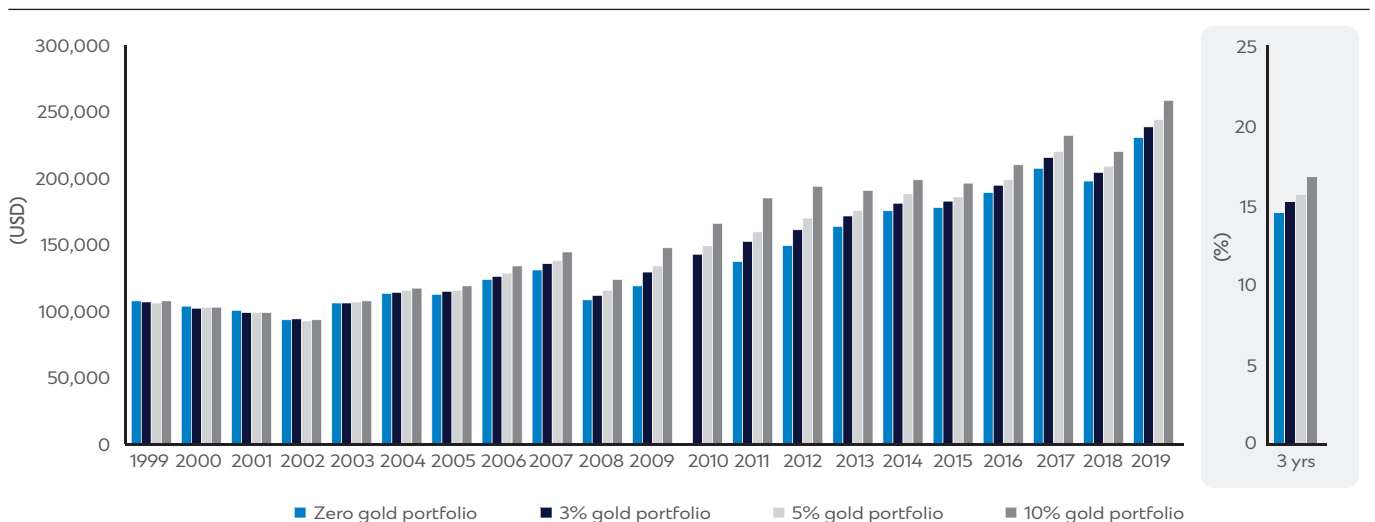


GoldSilver, a global bullion dealer, found that a traditional 60:40 portfolio of equity and bonds consistently underperformed portfolios that also had an allocation to gold over a 20-year period (1999-2019). This illustrates how gold can be an effective diversifier in an investment portfolio.

Separately, a portfolio with the highest gold allocation (10%) was the only one that raised the value of model portfolios from USD 100,000 to USD 250,000 over 20 years. A closer look at equity bear markets reveals that the portfolios with gold exposure, especially those with the higher exposures, had smaller drawdowns. This experiment highlights that holding gold as part of a portfolio across a range of market environments can optimise returns and mitigate risk. To deepen insights, this study was extended to assess gold's impact on pure equity portfolio returns (S&P 500). Over the three-year period ending 31 March 2026, even a modest 5% allocation to gold may have increased annual returns by nearly 1%. A 10% allocation could have added around 2% to returns. These results underscore gold's potential to elevate portfolio performance.

Fig. 3

Portfolio performance with and without gold



Source: GoldSilver, Standard Chartered; **RHS chart** - Investing.com (data for 31 March 2023 to 31 March 2026)

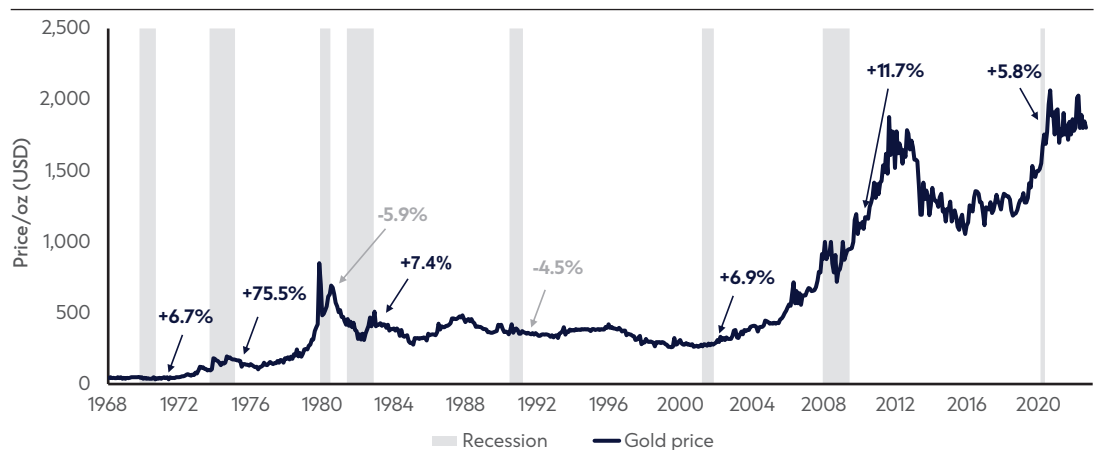
Safe haven in uncertain times

Gold during recession

About eight US recessions have been documented since 1973. In six of these, gold outperformed equities (S&P 500) by 37% on average. The exceptions were the 1981 and 1990 economic slowdowns. A deeper analysis of the 1981 crisis reveals that during this period, interest rates were aggressively hiked to combat the rising inflation. This made interest-bearing investments more attractive compared to gold. In contrast, the 1990 crisis occurred when most major central banks were net sellers of gold.

Fig. 4

Gold during recession



Source: Forbes, CME Group, Standard Chartered

Gold during stagflation

Stagflation is used to describe economic environments when growth stalls and the economy continues to suffer from high inflation. In the past, the US experienced stagflation in mid-to-late 1970s as high oil prices fuelled inflation and stalled economic growth. Typically, during periods of slowing growth, monetary stimulus and fiscal intervention can bring down borrowing costs and help businesses resume economic activities, eventually translating to sustained growth and lower unemployment. However, during stagflation, policy stimulus is constrained because it can risk worsening inflation.

Given this can result in very weak investor sentiment and low real (net-of-inflation) interest rates, investors tend to seek the safety of gold. During stagflation, any asset tied to the fiat currency or stock market faces stress. Investors seek safety, which is when gold emerges as a winner.

Fig. 5

Gold during stagflation

	Average real (inflation-adjusted) YoY total return since 1973, %					
	US equities	US govt. bonds	US T-bills	Commodities	Gold	REITs
Goldilocks	16.1	4.3	0.8	0.4	-2.5	18.1
Disinflation	8.4	8.1	1.7	-5.6	1.3	3.5
Reflation	14.6	-2.0	0.0	21.0	-1.1	14.0
Stagflation	-1.5	0.6	0.4	15.0	22.1	6.5

Source: Standard Chartered, Datastream Refinitiv and Schroders, Scottsdale Bullion & Coin. Data to 30 September 2021.
 Notes: Growth is proxied using US Conference Board Leading Economic Index and inflation is US core CPI. **Goldilocks** = accelerating growth and falling inflation; **Disinflation** = decelerating growth and inflation; **Reflation** = accelerating growth and inflation; **Stagflation** = decelerating growth and inflation above 10-year average.

Gold during crises

Gold can insulate a portfolio against extreme drawdowns. As can be seen in Figure 6, it outperformed the S&P 500 and other commodities in most past crisis events. Moreover, gold's low correlation with traditional assets such as stocks and bonds can be used to reduce overall portfolio volatility. This reduced dispersion has a stabilising effect during market drawdowns. Gold also does not carry credit risk, which makes it a confidence-inspiring portfolio constituent during periods of economic uncertainty when credit-related losses are higher than usual. In addition, gold is a highly liquid asset and does not have restrictions on withdrawals. These arguments help support gold's safe haven status.

Fig. 6

Gold during crisis

Crisis event	S&P 500	Gold (USD)	Commodities (USD)
Black Monday	-31.6	9.1	0.0
LTCM Crisis	-18.0	2.8	-2.8
Dot-com bubble	-23.1	-6.7	15.1
September 11 Terrorist attacks	-8.0	5.2	1.8
2002 recession	-31.7	7.8	1.7
Global Financial Crisis	-47.8	47.5	-36.1
Sovereign debt crisis v1	-11.1	6.4	-10.4
Sovereign debt crisis v2	-9.0	30.2	-2.1
2018 pullback	-19.6	6.6	-10.1
COVID market crash	-20.0	6.2	-23.5
Average return	-22.0	11.5	-6.6

Source: Standard Chartered, First Links, Morningstar

Methods to invest in gold

Gold investment routes can be broadly segmented into physical and virtual approaches.

Physical gold



- **Gold bullion** refers to gold bars that are traded on a commodity exchange. Typically, bars ranging from approximately 10 ounces to 400 ounces are traded. Fractional gold bars are not traded and, hence, the minimum capital required to invest in bullion is relatively high. Insurance and proper storage become essential costs when holding such physical assets.
- **Gold coins** start at 1 ounce or lower; there are also a range of collectible gold coins such as Canadian Maple leaf coins (99.9% purity) and South African Krugerrands (the modern world's first gold bullion coin). These coins do not always reflect prevalent gold prices and may be driven by very specific demand-supply dynamics.
- WGC data indicate that **gold jewellery** accounts for almost 50% of the gold mined worldwide. The markup on jewellery is determined based on multiple factors such as labour cost, material cost, overheads, inflation and demand-supply dynamics. Jewelry Mavericks, a jewellery business consultant, estimates the average markup on jewellery to be around 50-100%. Furthermore, gold jewellery may not always use a similar level of purity as bullion or coins.

VS

Virtual gold



- **Gold funds** invest in gold assets that can include gold bullion, gold mining companies and gold ETFs. This approach is convenient, can be achieved with relatively small amounts of capital and aims to provide exposure to gold and its related diversification and inflation hedge characteristics. However, the performance may not always align with raw gold given costs associated with funds and the manner in which a particular fund chooses to take exposure. This approach offers better liquidity compared to physical gold.
- **Gold ETFs** usually hold physical gold bullion as their underlying asset. ETFs tend to track physical gold's price more intricately than funds both due to their lower costs and more focussed underlying holdings. They also offer better liquidity as they can be traded intra-day but therefore carry higher intra-day volatility risk.
- **Gold-related stocks** are an indirect way to gain exposure to gold. Usually these are stocks of companies that are engaged in the exploration and production of gold. These stocks are influenced by not only gold prices, but also the operational performance and idiosyncratic risks of such companies, as well as broader equity market risks. Compared with other virtual investing options, this method may be subject to additional risks such as regulatory changes and environmental concerns.
- **Gold futures and options** are financial derivatives that derive their value from raw gold. These tend to be high risk and are usually traded by investors with a deep understanding of derivatives markets.

Private access to gold market

If you want to buy, sell and securely store gold outside of public exchanges, keep transactions discreet and holdings confidential, private access to the gold market might just be for you.

This approach is especially attractive to affluent investors seeking privacy, direct control and ownership of their gold assets. Furthermore, private gold markets provide global exposure through offshore vaulted gold, gold-backed tokens with worldwide trading access and allocations managed by international family offices.

In addition to investing in bullions and coins through direct transactions with miners and dealers, there are other investment opportunities within the private gold markets that you can consider exploring.

Investing in **vaulted gold** involves purchase of large quantities of physical gold (weights over 1 kg). Unsurprisingly, this means additional requirements such as secure storage, professional custodians, insurance, audits and vault services. However, these services also drive up costs, which is why this mode of investing tends to have high minimum investment thresholds, limiting access to affluent customers.

Private gold funds are investment vehicles, similar to mutual funds, that pool capital from affluent investors to invest in gold assets. However, their structure, strategy and access differ from public gold ETFs or mutual funds. Private gold funds can be structured as:

- **Hedge funds**, which can use techniques such as short selling, leveraging or derivative trading (gold futures and options) to generate outsized returns. This flexible structure also offers tax benefits and the ability to create bespoke contracts tailored to the investor's preferences.
- **Alternative investment funds**, which pool investor capital in gold or related assets and are managed by professional fund managers operating under comparatively

lighter regulations. This allows for greater flexibility in investment strategy, asset selection and use of leverage. Importantly, offshore fund structures can further offer global exposure and potential tax efficiencies.

- Private gold funds structured as **family office vehicles are similar to trusts**: they invest the wealth of a single family in gold assets. This structure offers improved privacy, tax efficiency, greater control and certain regulatory exemptions, making it particularly well-suited for affluent investors seeking long-term, customised exposure to gold.

Gold mining ventures are high-risk, high-reward investment options where investors provide capital to companies engaged in exploration, development or expansion of gold mines. In return, depending on the terms of the agreement, investors may receive equity (ownership stakes), convertible debt or royalty interests. These deals are typically structured as private placements, venture capital or private equity-style arrangements and may require substantial minimum investments. Where such investments become tricky is their high exposure to a broad range of operational, environmental, political and geographic risks, in addition, of course, to fluctuations in gold prices.

As is increasingly clear, the private route to investing in gold is suitable for sophisticated investors seeking broad diversification and likely outsized returns but willing to take on significantly higher risk. Over time, there has been a clear shift among investors toward creating a positive impact while also generating financial returns. As a result, sustainable ways of investing in gold are increasingly becoming top of mind for investors.

Sustainable gold investing

Sustainability in gold investing is rapidly gaining traction worldwide, driven by regulatory requirements, growing environmental concerns and increasing investor demand for ethical business practices. In 2025, sustainable practices in global gold production increased by over 40%, underscoring significant progress in environmental responsibility and supply chain transparency.

Several key factors play a crucial role when evaluating the impact, credibility and long-term value of sustainable investing in gold.

- Ethical sourcing of gold refers to mining, production and trading practices that respect human rights, protect the environment and ensure fair working conditions. Companies committed to transparency and ethical practices often carry certifications such as Fairtrade or Fairmined (safe conditions and fair pay). Similarly, LBMA Responsible Gold Guidance ensures that gold is mined in ways that uphold labour and environmental standards. Investors can check for trusted certifications to confirm that the gold is indeed being sourced responsibly.
- Investors who wish to consider sustainability in their gold investment strategy can prioritise companies that adhere to sustainable mining practices. Typically, these companies use cleaner technologies, adhere to ethical sourcing standards and maintain transparent ESG reporting. Investors can evaluate a company's alignment with ESG standards. Reports and ratings from firms such as Sustainalytics or Moody's show how the company handles operational practices such as water use, labour practices and carbon emissions.
- Sustainable gold investing supports a more resilient portfolio as it helps in identifying companies committed to responsible practices and long-term risk management. VanEck, a global investment management firm, incorporates ESG considerations into its gold investment process. They found that gold mining companies with strong ESG standards tend to be more transparent, better managed and less prone to regulatory risks.
- Impact gold products such as gold funds/ETFs with ESG overlay and recycled gold (gold reused from old products such as electronics, watches and dental alloys) help in achieving financial returns alongside positive environmental or social outcomes. Investing in impact gold products support value-driven investing goals and are, therefore, appealing to investors who seek to profit with a purpose.

Sustainability considerations can serve as a strategic edge to traditional gold investing by adding the humane aspect without compromising on the core benefits of financial returns, inflation hedging and diversification. A key reason for investing in gold is its ability to preserve and transfer wealth across generations.

Gold in legacy planning

Gold is a widely recognised asset in legacy planning owing to its stability, liquidity and enduring value. In many Asian cultures, it also holds deep emotional and cultural significance across generations.

Gold needs special attention in estate planning to ensure seamless transfer, avoid legal complications and protect the interest of heirs, because:

- Unlike real estate or vehicles, gold may lack formal deeds or registries; clear instructions in an estate plan are essential to streamline inheritance
- Explicit instructions in an estate document can prevent family disputes and lengthy probate proceedings. This will also ensure that gold is handed down as per the owner's wishes and not left to legal distribution
- Thorough planning helps in optimising inheritance tax, ensuring compliance with regional regulations and protecting the value of assets for heirs.

With the importance of thorough legacy planning for gold underlined, we look at a few practical steps to streamline the process.



Steps for gold legacy planning

Inventory and documentation

Maintain a comprehensive inventory of your gold holdings to gain clear visibility into exactly what you own, ensure accurate records for insurance and valuation purposes, and facilitate a more organised transfer of assets to heirs.

- For physical gold, record ownership details, purchase dates, quantities, purity, market value, condition and precise storage locations.
- For digital gold, document account ownership, transaction records and access instructions.

Update nominations on your digital gold holdings

This will ensure rightful transfer, avoid delays, protect asset value and reduce legal risks.

Optimise taxes

Consider legal ways to optimise on estate taxes. For instance, gold gifting during the owner's lifetime may be tax-exempt in some jurisdictions. Check for local laws to use them favourably.

Review and reassess regularly

Review and adjust the legacy plan periodically to align with changing gold holdings, family circumstances and legal environment.

Step
01

Step
02

Step
03

Step
04

Step
05

Step
06

Step
07

Include gold in estate documents

Include clear instructions in wills or trusts detailing how gold holdings (physical and digital) should be distributed among heirs, specifying beneficiaries, asset allocation and any preferred methods for transferring or distributing the gold. Clear documentation helps to ensure that the process is straightforward and minimises disputes.

Engage an estate planning expert

Guidance from an estate planning attorney can help avoid conflicts, manage tax implications and align the gold assets with your overall legacy goals.

Inform and educate heirs

Open communication is an important part of the legacy process. Informing and educating heirs can help in avoiding conflicts, delays or legal issues. It can also foster responsible stewardship, especially in case of generational gold.

A well-planned gold legacy ensures that its long-term benefits, such as portfolio resilience and inflation hedging, are preserved and transferred without any hassle.

Conclusion

Investments in gold can be an attractive way to strengthen an investment portfolio's resilience and wealth preserving capabilities.

The low correlation of gold with traditional assets like equities and bonds means that it works well as a portfolio diversifier in addition to being a safe haven asset during periods of high volatility or economic uncertainty. History shows that the yellow metal has the potential to create and preserve wealth over the long haul.



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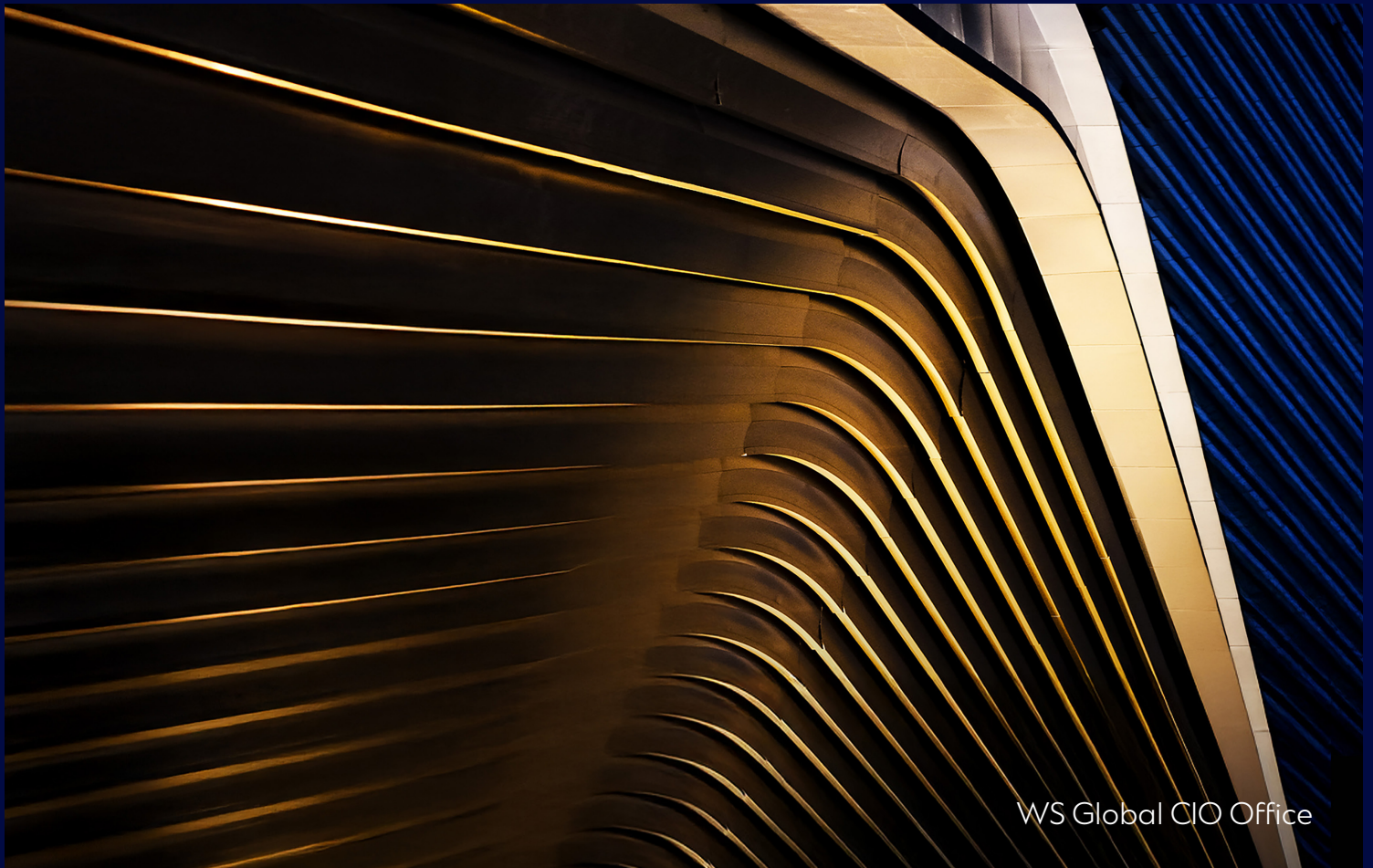
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Gold market dynamics

Revisiting the safe haven

April 2026



WS Global CIO Office

Gold's relevance in modern portfolios extends beyond its historical status as a store of value.

In an environment marked by recurring geopolitical frictions, macroeconomic fragmentation and shifting policy regimes, gold has demonstrated an enduring capacity to act as a long term portfolio stabiliser, lending resilience when other assets face synchronised drawdowns. This time-tested attribute continues to underpin its relevance in diversified investment strategies.

In this report, we explore gold's role as a hedge against market volatility and economic instability. We also synthesise insights from historical analyses and review gold's performance across market cycles. Furthermore, we discuss the various methods to include gold in a portfolio.

Overview of gold

The World Gold Council (WGC) estimated that the stock of gold above ground was approximately 220,000 tonnes as of 2025. Based on March-end 2026 gold prices, the stock's value stood at around USD 33trn, nearly 53% higher than a year ago, illustrative of the sharp rise in gold prices recently.

From an end-user perspective, gold remains a cornerstone in making jewellery and ornaments and showcasing wealth, which continue to be key demand drivers for this asset class. Historically, gold has been viewed as a currency, largely due to its role as a store of value. Gold is also increasingly incorporated in electrical equipment for longevity and improved performance. Importantly, however, gold has found its way into investment portfolios given its perceived quality of being an inflation hedge over long periods and its low correlation with equity and bonds.



Gold

- Known for stability, reliability
- Acts as an inflation hedge
- Holds low correlation with stocks and bonds (long term)
- Seen as a safe haven during a crisis

Source: Standard Chartered

Steve Brice
Global Chief Investment Officer

Manpreet Gill
Chief Investment Officer, Africa, Middle East & Europe (AMEE)

Why consider gold?

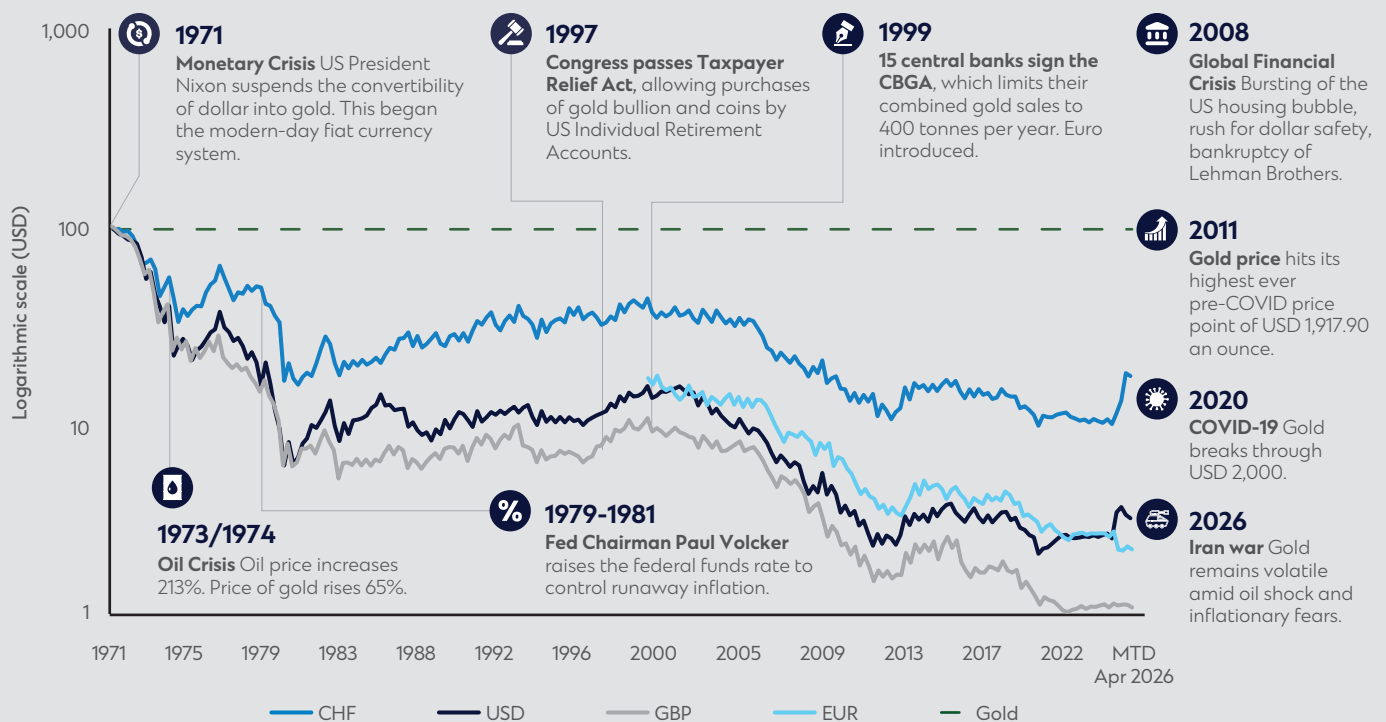
In this section, we examine how good gold is as an inflation hedge and a portfolio diversifier, and whether it deserves its safe haven status.

Inflation hedge

WGC data suggest that, over long periods (40+ years), gold can be a good inflation hedge. During inflationary periods, the purchasing power of fiat currency declines, but gold tends to hold, or grow, in value. Gold's naturally limited supply is an additional support. The inverse relationship between gold and major fiat currencies has held since the 1970s.

Fig. 1

Major fiat currencies vs gold



Source: Standard Chartered

One of the most inflationary periods in recent history was between 1973 and 1979, when average annual inflation in the US was around 8.2%. This was, of course, led by a jump in oil prices. During this period, gold gained almost 35% annually, an outcome supporting its status as a hedge against inflation. A recent study by WGC, analysing data from 1971 to 2020, found that gold delivered an annual growth rate of 8% to 10%, significantly outpacing CPI inflation of approximately 3.9%. Gold's value as a financial asset ties it more closely to economic growth and monetary expansion than to CPI alone, reinforcing its role as a long-term inflation hedge with positive real returns.

Note that while gold's performance was strong when inflation was high, some of this performance subsequently reversed once inflation was tamed via sharply higher interest rates as a result of tightening Fed policy. Once interest rates were raised sharply to bring inflation under control, gold failed as an effective hedge against inflation. For instance, between 1975 and 1985, a period of tightening US monetary policy stance, gold experienced five instances of negative annualised returns, with the largest such decline exceeding 30% in USD terms in 1981.

A more recent inflationary period was in 2021-22, when prices shot up as a result of COVID-led supply chain disruption, a jump in government spending and relatively low interest rates. US inflation hit a high of over 9% in June 2022. Gold hit a cyclical high of around USD 2,050 in March 2022, supporting the view that it acts as a hedge against inflation. However, the reverse also held – gold subsequently fell over 20% and did not bottom until October 2022. This fall in gold prices coincided with the Fed's aggressive interest rate hiking cycles. Higher bond yields (especially net of inflation) and a stronger USD are two factors that worked against gold – both in this instance and over longer periods of time. What this shows is that while gold remains an attractive hedge against rising inflation, tightening Fed policy aimed at taming inflation can hurt gold prices.

A similar paradox emerged in early 2026 during the Iran war, when gold hit an all time high of USD 5,589 in January 2026, before crashing nearly 25% by March 2026 to approximately USD 4,130, and finally rebounding to its long term uptrend. Despite geopolitical tensions that had driven safe-haven demand, oil shock, inflation fears and expectations of higher rates pressured gold downward.

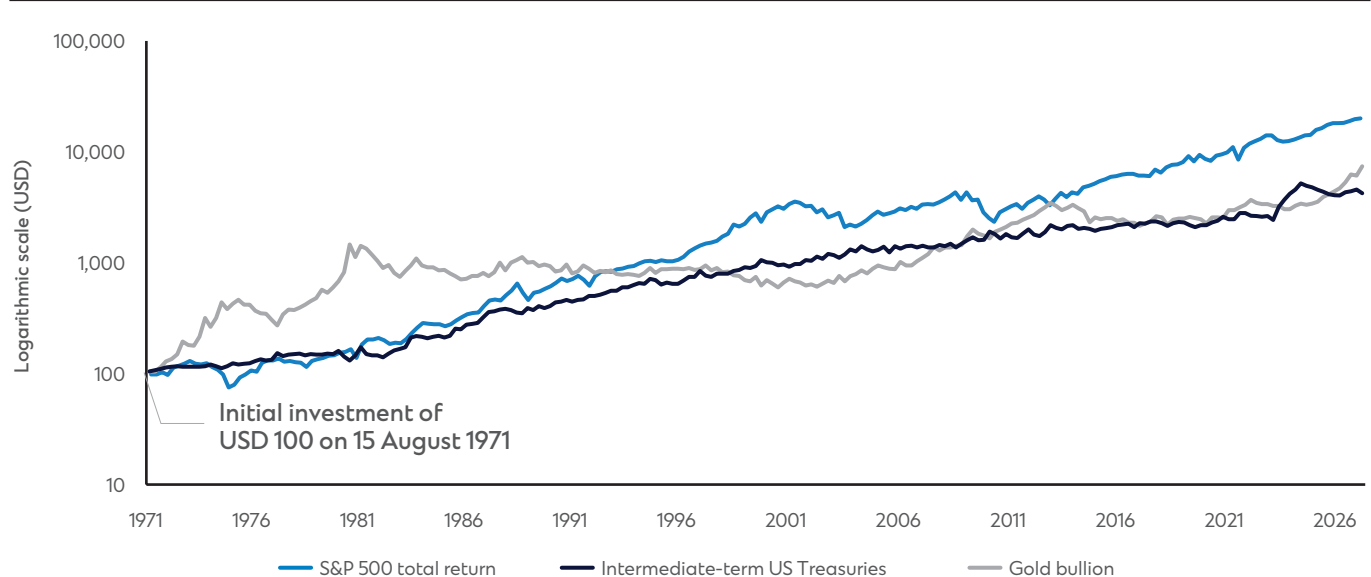
The aforesaid instances of gold's rise and subsequent fall indicate that gold may not be the 'perfect' safe haven at all times and that it too can succumb to the vagaries of economic and market cycles. However, over the past 50 years, the average annual rate of return from gold has been approximately 8%. Such a performance suggests that while gold can face temporary headwinds, it can potentially provide inflation-beating returns in the long term.

Portfolio diversification

Over the past 50+ years, gold has shared a low correlation with equities. Meanwhile, gold's correlation with US government bonds is generally negative during periods of high inflation, as bond prices tend to fall but gold prices rise during such periods. During crises, however, markets instead view both gold and US government bonds as safe haven assets, resulting in a slightly positive correlation between them. Gold's overall correlation with bonds in the long term, however, is low.

Fig. 2

Gold, bonds and equities



Source: TopForeignStocks.com, Standard Chartered

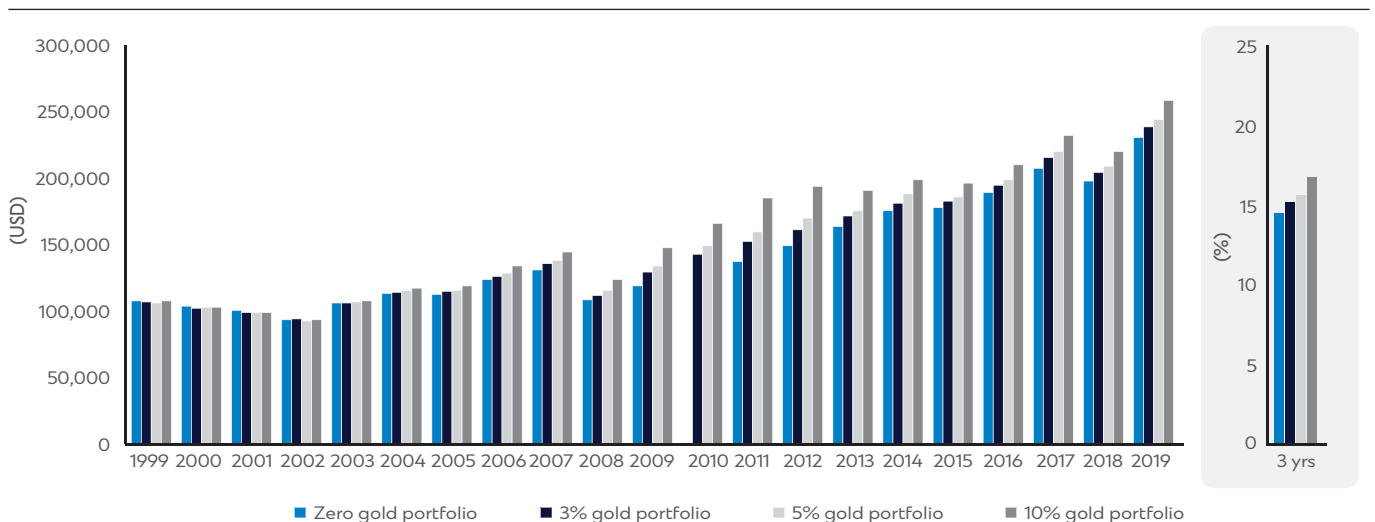


GoldSilver, a global bullion dealer, found that a traditional 60:40 portfolio of equity and bonds consistently underperformed portfolios that also had an allocation to gold over a 20-year period (1999-2019). This illustrates how gold can be an effective diversifier in an investment portfolio.

Separately, a portfolio with the highest gold allocation (10%) was the only one that raised the value of model portfolios from USD 100,000 to USD 250,000 over 20 years. A closer look at equity bear markets reveals that the portfolios with gold exposure, especially those with the higher exposures, had smaller drawdowns. This experiment highlights that holding gold as part of a portfolio across a range of market environments can optimise returns and mitigate risk. To deepen insights, this study was extended to assess gold's impact on pure equity portfolio returns (S&P 500). Over the three-year period ending 31 March 2026, even a modest 5% allocation to gold may have increased annual returns by nearly 1%. A 10% allocation could have added around 2% to returns. These results underscore gold's potential to elevate portfolio performance.

Fig. 3

Portfolio performance with and without gold



Source: GoldSilver, Standard Chartered; **RHS chart** - Investing.com (data for 31 March 2023 to 31 March 2026)

Safe haven in uncertain times

Gold during recession

About eight US recessions have been documented since 1973. In six of these, gold outperformed equities (S&P 500) by 37% on average. The exceptions were the 1981 and 1990 economic slowdowns. A deeper analysis of the 1981 crisis reveals that during this period, interest rates were aggressively hiked to combat the rising inflation. This made interest-bearing investments more attractive compared to gold. In contrast, the 1990 crisis occurred when most major central banks were net sellers of gold.

Fig. 4

Gold during recession



Source: Forbes, CME Group, Standard Chartered

Gold during stagflation

Stagflation is used to describe economic environments when growth stalls and the economy continues to suffer from high inflation. In the past, the US experienced stagflation in mid-to-late 1970s as high oil prices fuelled inflation and stalled economic growth. Typically, during periods of slowing growth, monetary stimulus and fiscal intervention can bring down borrowing costs and help businesses resume economic activities, eventually translating to sustained growth and lower unemployment. However, during stagflation, policy stimulus is constrained because it can risk worsening inflation.

Given this can result in very weak investor sentiment and low real (net-of-inflation) interest rates, investors tend to seek the safety of gold. During stagflation, any asset tied to the fiat currency or stock market faces stress. Investors seek safety, which is when gold emerges as a winner.

Fig. 5

Gold during stagflation

	Average real (inflation-adjusted) YoY total return since 1973, %					
	US equities	US govt. bonds	US T-bills	Commodities	Gold	REITs
Goldilocks	16.1	4.3	0.8	0.4	-2.5	18.1
Disinflation	8.4	8.1	1.7	-5.6	1.3	3.5
Reflation	14.6	-2.0	0.0	21.0	-1.1	14.0
Stagflation	-1.5	0.6	0.4	15.0	22.1	6.5

Source: Standard Chartered, Datastream Refinitiv and Schroders, Scottsdale Bullion & Coin. Data to 30 September 2021. Notes: Growth is proxied using US Conference Board Leading Economic Index and inflation is US core CPI. **Goldilocks** = accelerating growth and falling inflation; **Disinflation** = decelerating growth and inflation; **Reflation** = accelerating growth and inflation; **Stagflation** = decelerating growth and inflation above 10-year average.

Gold during crises

Gold can insulate a portfolio against extreme drawdowns. As can be seen in Figure 6, it outperformed the S&P 500 and other commodities in most past crisis events. Moreover, gold's low correlation with traditional assets such as stocks and bonds can be used to reduce overall portfolio volatility. This reduced dispersion has a stabilising effect during market drawdowns. Gold also does not carry credit risk, which makes it a confidence-inspiring portfolio constituent during periods of economic uncertainty when credit-related losses are higher than usual. In addition, gold is a highly liquid asset and does not have restrictions on withdrawals. These arguments help support gold's safe haven status.

Fig. 6

Gold during crisis

Crisis event	S&P 500	Gold (USD)	Commodities (USD)
Black Monday	-31.6	9.1	0.0
LTCM Crisis	-18.0	2.8	-2.8
Dot-com bubble	-23.1	-6.7	15.1
September 11 Terrorist attacks	-8.0	5.2	1.8
2002 recession	-31.7	7.8	1.7
Global Financial Crisis	-47.8	47.5	-36.1
Sovereign debt crisis v1	-11.1	6.4	-10.4
Sovereign debt crisis v2	-9.0	30.2	-2.1
2018 pullback	-19.6	6.6	-10.1
COVID market crash	-20.0	6.2	-23.5
Average return	-22.0	11.5	-6.6

Source: Standard Chartered, First Links, Morningstar

Methods to invest in gold

Gold investment routes can be broadly segmented into physical and virtual approaches.

Physical gold



- **Gold bullion** refers to gold bars that are traded on a commodity exchange. Typically, bars ranging from approximately 10 ounces to 400 ounces are traded. Fractional gold bars are not traded and, hence, the minimum capital required to invest in bullion is relatively high. Insurance and proper storage become essential costs when holding such physical assets.
- **Gold coins** start at 1 ounce or lower; there are also a range of collectible gold coins such as Canadian Maple leaf coins (99.9% purity) and South African Krugerrands (the modern world's first gold bullion coin). These coins do not always reflect prevalent gold prices and may be driven by very specific demand-supply dynamics.
- WGC data indicate that **gold jewellery** accounts for almost 50% of the gold mined worldwide. The markup on jewellery is determined based on multiple factors such as labour cost, material cost, overheads, inflation and demand-supply dynamics. Jewelry Mavericks, a jewellery business consultant, estimates the average markup on jewellery to be around 50-100%. Furthermore, gold jewellery may not always use a similar level of purity as bullion or coins.

VS

Virtual gold



- **Gold funds** invest in gold assets that can include gold bullion, gold mining companies and gold ETFs. This approach is convenient, can be achieved with relatively small amounts of capital and aims to provide exposure to gold and its related diversification and inflation hedge characteristics. However, the performance may not always align with raw gold given costs associated with funds and the manner in which a particular fund chooses to take exposure. This approach offers better liquidity compared to physical gold.
- **Gold ETFs** usually hold physical gold bullion as their underlying asset. ETFs tend to track physical gold's price more intricately than funds both due to their lower costs and more focussed underlying holdings. They also offer better liquidity as they can be traded intra-day but therefore carry higher intra-day volatility risk.
- **Gold-related stocks** are an indirect way to gain exposure to gold. Usually these are stocks of companies that are engaged in the exploration and production of gold. These stocks are influenced by not only gold prices, but also the operational performance and idiosyncratic risks of such companies, as well as broader equity market risks. Compared with other virtual investing options, this method may be subject to additional risks such as regulatory changes and environmental concerns.
- **Gold futures and options** are financial derivatives that derive their value from raw gold. These tend to be high risk and are usually traded by investors with a deep understanding of derivatives markets.

Conclusion

Investments in gold can be an attractive way to strengthen an investment portfolio's resilience and wealth preserving capabilities.

The low correlation of gold with traditional assets like equities and bonds means that it works well as a portfolio diversifier in addition to being a safe haven asset during periods of high volatility or economic uncertainty. History shows that the yellow metal has the potential to create and preserve wealth over the long haul.



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360 Perspective

Private Infrastructure

April 2026

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WS Global CIO Office

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Steve Brice
Global Chief Investment Officer

Hannah Chew
Portfolio Strategist



What are Infrastructure Assets?

Infrastructure assets are the networks and systems that provide essential services, facilitate economic activity and enable the movement or storage of goods, water, energy, data and people



Traditional Infrastructure

Upgrades on roads, rails, transport network (e.g., ports, waterways and airports); modernizing public transit; build, preserve and retrofit civic and commercial real estate; upgrading aging water infrastructure including water treatment plants, pipes, sewers, water sanitation, improving access to clean water.



Renewable Infrastructure

Investments to accelerate the adoption of renewables in primary energy supply to meet Paris agreement commitments: Upgrading the electric grid, building electric vehicle charging stations, supporting sustainable farming and circular economy initiatives, decarbonizing the energy sector.



Digital Infrastructure

Investments in digitalization initiatives (cloud, 5G, data centers), expansion of high-speed broadband, funding of scientific research, granting or improving digital access to underserved segments of the population.

Source: Standard Chartered



Key characteristics of infrastructure assets

Key characteristics

Summary

'Real' Asset Class

- Infrastructure represents a tangible asset – i.e., a combination of land and structures that constitutes real property. Real property is expected to almost always retain a residual value, which is particularly attractive during periods of distress

Essential Services

- Given the essential nature of Infrastructure services, user demand tends to be relatively inelastic. Therefore, they tend to exhibit a lower correlation to the economic cycle compared with other sectors

High Barriers of Entry

- Infrastructure requires a high level of initial capital investment, which acts as a significant impediment to potential competitors entering the market. Assets can enjoy monopolistic or quasi-monopolistic market positioning

High Regulation

- Infrastructure assets are usually essential services providers which are important in the function of society and economy. Therefore, governments tend to exercise strict regulation over them. While regulations can vary, they usually make cash flows more stable

Diversified end-user base

- The counterparties to infrastructure assets are generally a widely diversified group of end users, which helps stabilize cash flows

Cash flow predictability

- Regulation provides long-term revenue visibility to infrastructure investments

Inflation hedge

- Depending on the type of infrastructure asset, price inflation can sometimes be passed on to the end consumer. Most regulatory frameworks allow regulated assets to use inflation-indexed user tariffs

Key
characteristics
of infrastructure
assets

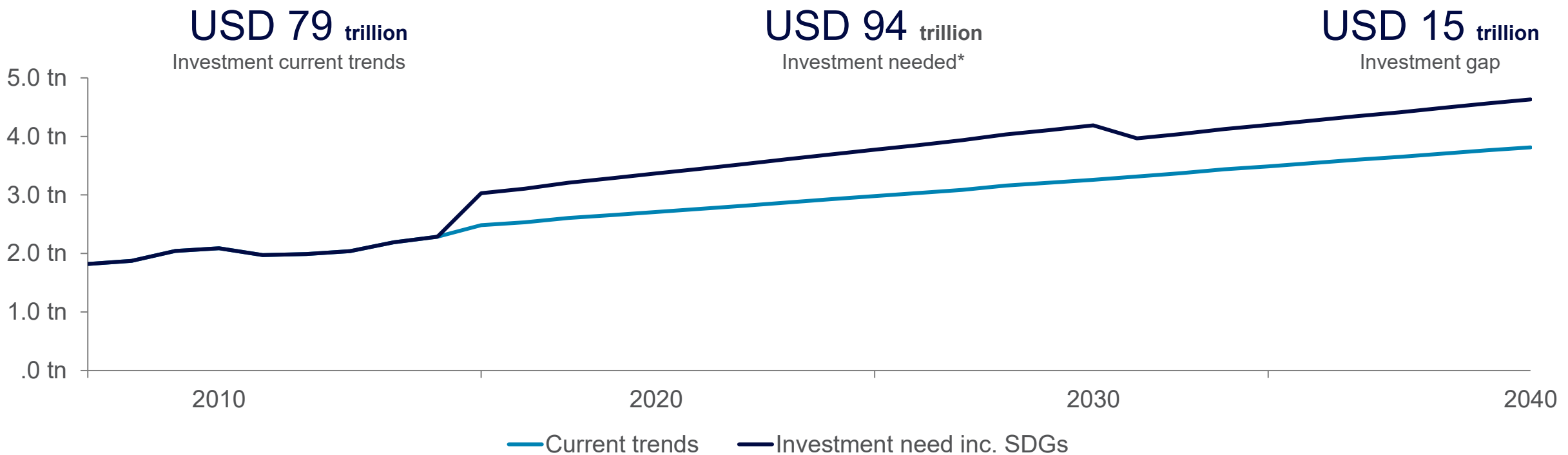
Source: Deutsche Asset Management, Standard Chartered



The Infrastructure “super-cycle”

Post-COVID, major governments are investing heavily in infrastructure assets to achieve supply chain resilience, digital independence and environmental targets

Global infrastructure investment spend has trailed behind actual needs



Source: Global Infrastructure Outlook, Standard Chartered
*Includes the additional investment needed for countries that have not yet met the SDGs
Note: values refer to estimated investment spending and needs until 2040
SDG refer to United Nations Sustainable Development Goals

The definition of infrastructure is evolving to meet future needs

From traditional, asset-heavy sectors...

Asset-heavy

Large; physical, high-capex-assets

Highly regulated

Government-owned and monopoly models

Linear and centralized

Centralized grids, fixed networks

Capex-intensive

Heavy investments upfront

Built on long-established technologies

Fossil fuels, resource-heavy

... to inclusion of services, technology, and distributed systems



Tech-enabled

Digital, software-driven infrastructure as a service



Market-driven

Private-sector competition and digital platforms



Decentralized and modular

Distributed, flexible, and demand driven



Opex and services

Pay-as-you-go, cloud-based subscription models

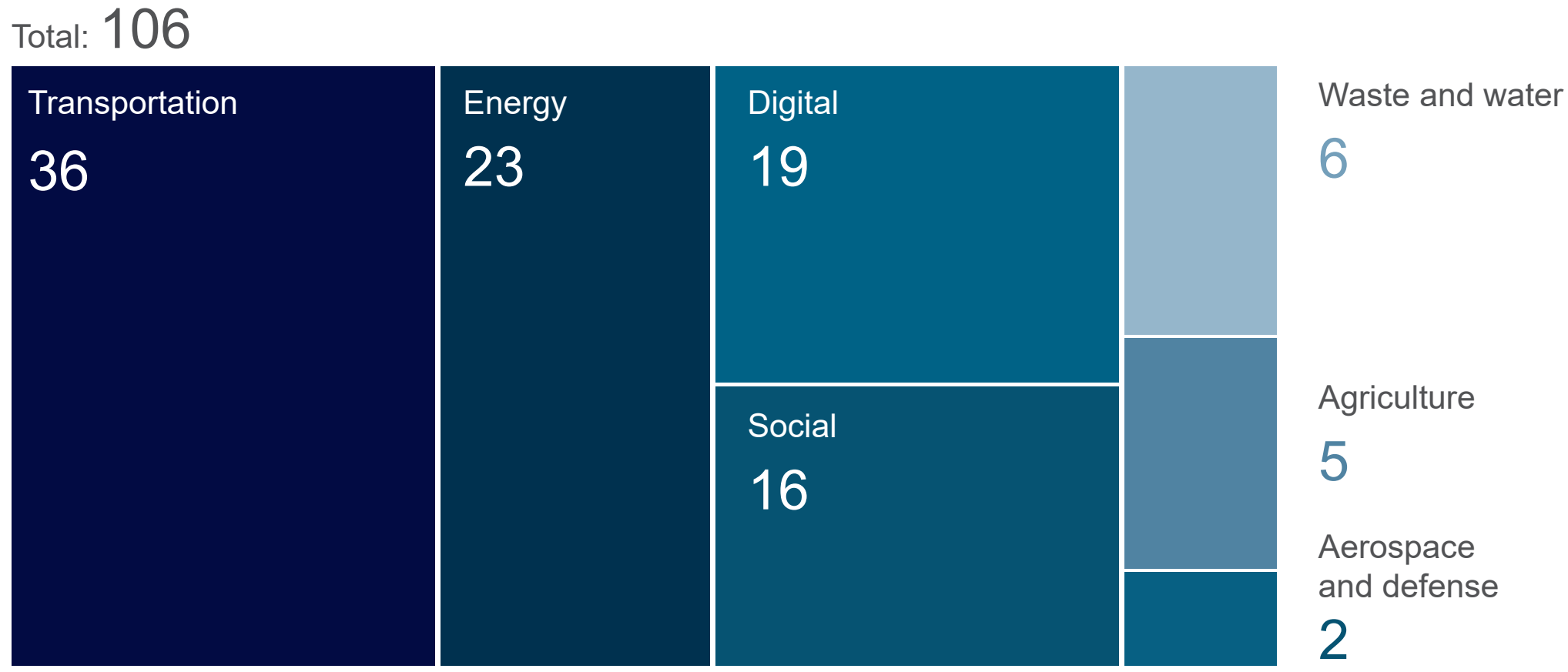


Built on scaling and disruptive technologies

Low-carbon, circular economy, sustainability driven

Cumulative infrastructure investment is expected to reach as high as \$106 trillion by 2040

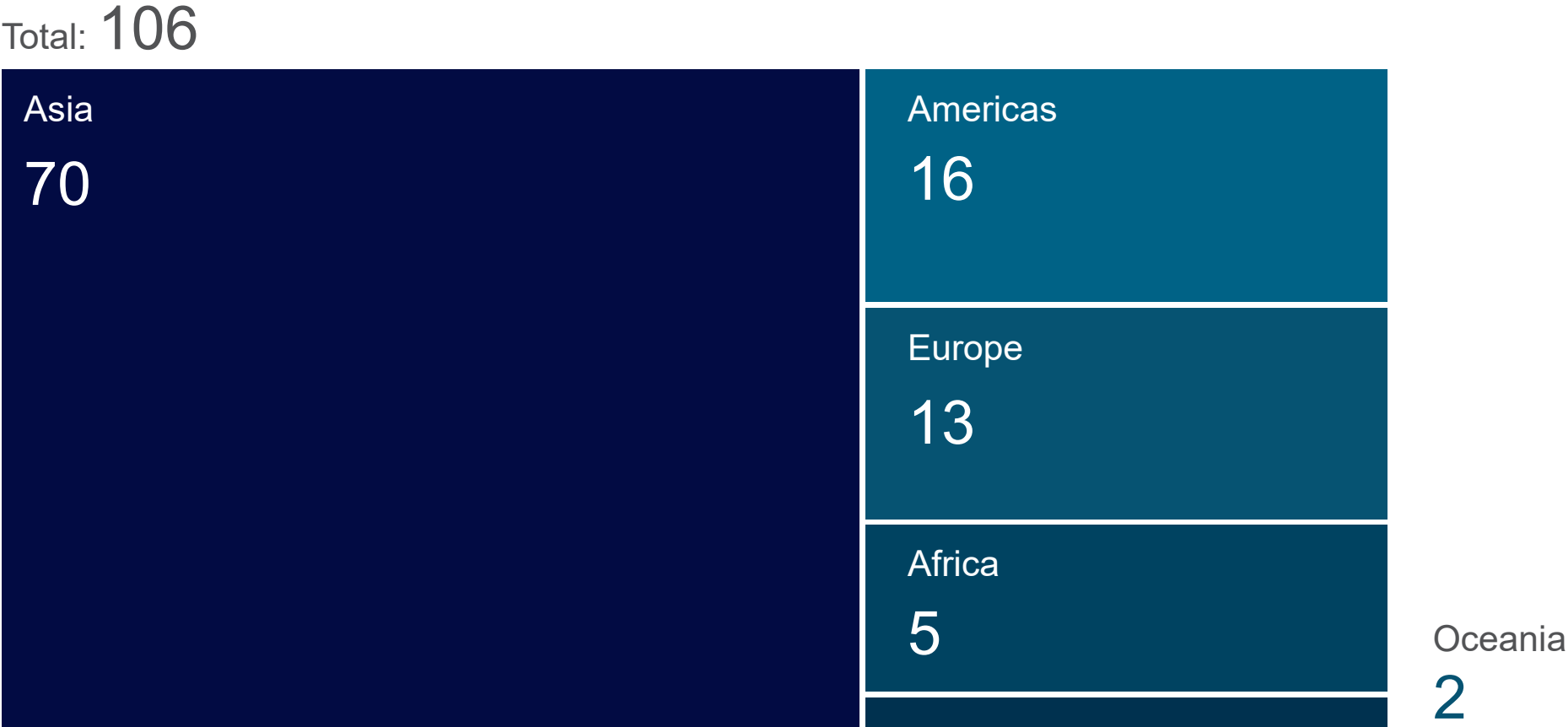
Total infrastructure investment projected through 2040, by sector, \$ trillion



Source: Food and Agriculture Organization; Global Infrastructure Hub; International Energy Agency; International Monetary Fund; Organisation for Economic Co-operation and Development; Preqin; United Nations; World Bank; World Economic Forum; McKinsey
Note: Figures do not sum, because of rounding

Asia could receive two-thirds of the total infrastructure investment through 2040

Total infrastructure investment projected through 2040, by region, \$ trillion



Source: Food and Agriculture Organization; Global Infrastructure Hub; International Energy Agency; International Monetary Fund; Organisation for Economic Co-operation and Development: Preqin; United Nations; World Bank; World Economic Forum; Mckinsey



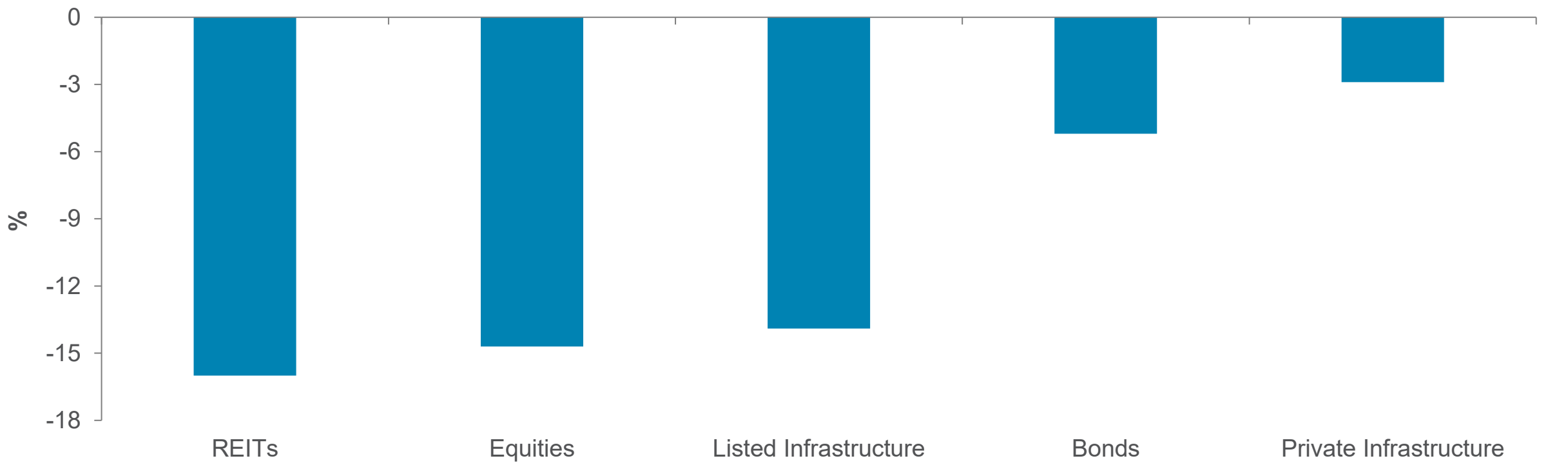


Portfolio Benefits of Private Infrastructure

Private infrastructure assets have demonstrated resilient returns during drawdowns

History suggests that infrastructure assets tend to perform relatively well compared to traditional assets during worst performing quarters for equity markets

Average return by asset class during 10 worst quarters for equity markets



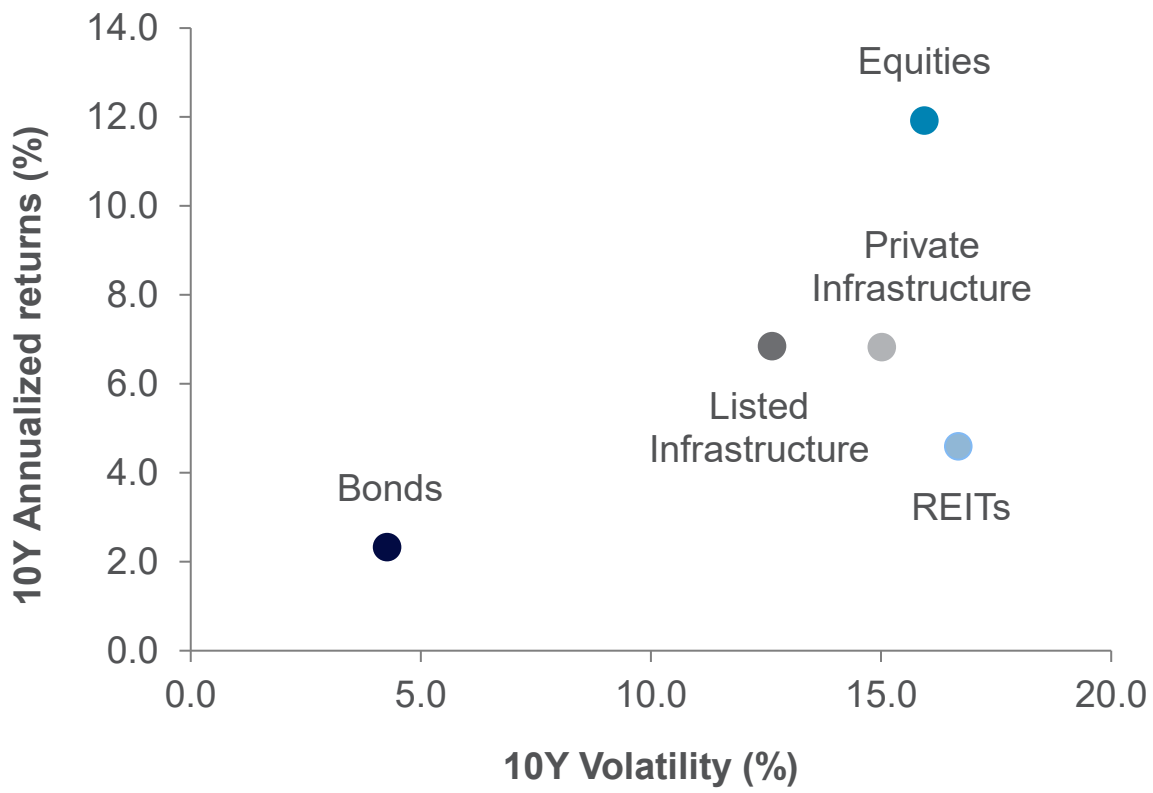
Source: Allianz Research, Standard Chartered



Private infrastructure is often less correlated to traditional asset classes

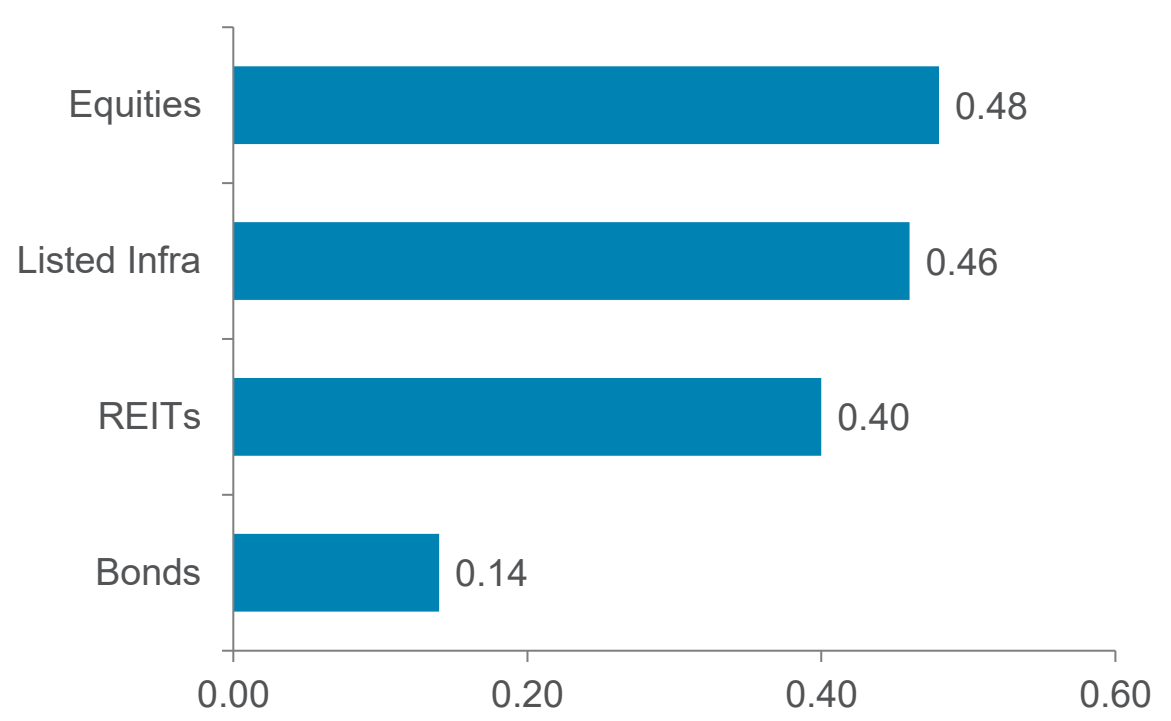
It also has relatively lower volatility compared to equities and REITs, given its regular cash flow which can mitigate downside price volatility

10Y Risk/Return matrix across Asset Class



Source: Bloomberg, The Scientific Infrastructure & Private Assets Research Institute (EIPA), Standard Chartered Bank
Data as of September 2025

10Y Correlation with Private Infrastructure



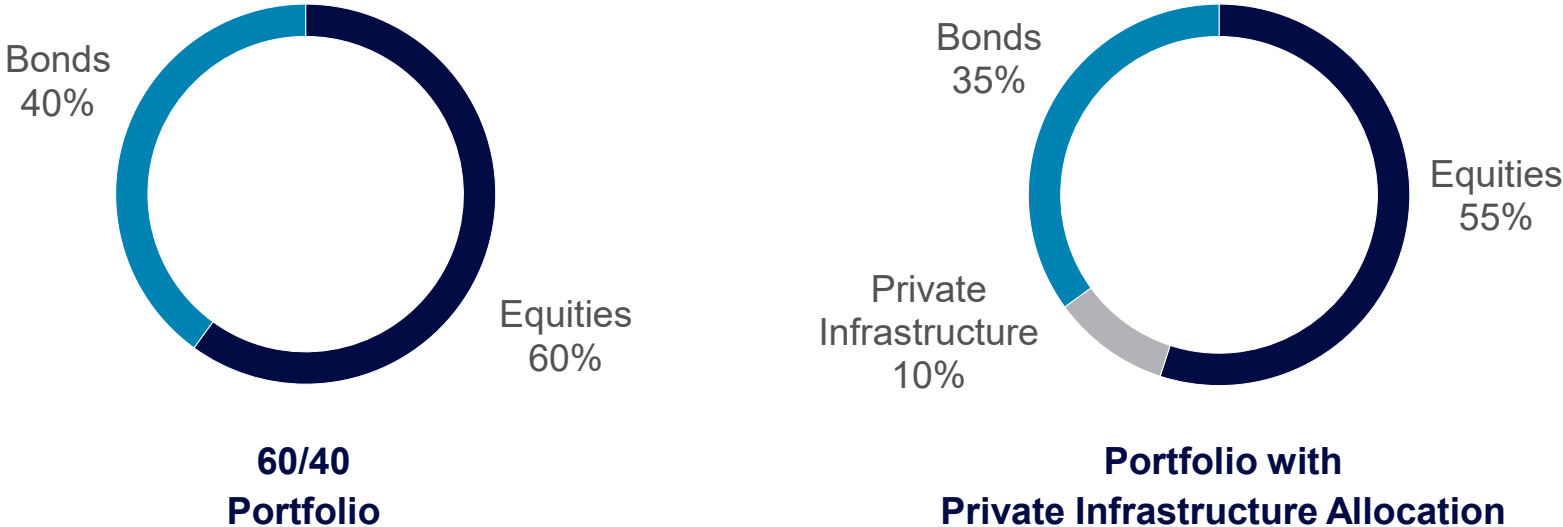
Source: Allianz Research, Standard Chartered Bank
Data as of 2024



Adding Private infrastructure could improve risk-adjusted returns in a portfolio

An allocation towards Private Infrastructure, which is a less volatile asset, can improve a portfolio's risk-adjusted returns

Performance of a portfolio with a private infrastructure allocation vs a Traditional 60/40 Portfolio



	60/40 Portfolio	Portfolio with Private Infrastructure Allocation
Historical 10Y Ann. Return	8.7%	9.0%
Historical Volatility	11.1%	10.3%
Sharpe Ratio	0.47	0.54

Source: StashAway, Standard Chartered
Data as of September 2024



Infrastructure assets are generally resilient through an economic cycle

Private Infrastructure assets can help hedge against inflation as cash flows are typically inflation-linked, which is a valuable attribute during periods of high inflation



Income

Given their essential nature, infrastructure investments can generate relatively stable and resilient cash-flows in a range of market environments



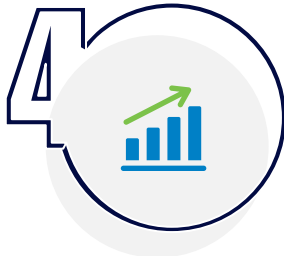
Downside Protection

The essential nature of the business allows it to remain resilient through an economic cycle, usually limiting drawdowns at least relative to public market assets



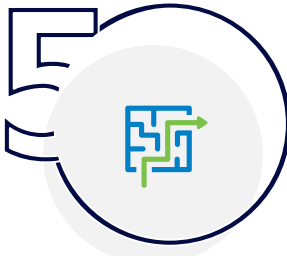
Stability of Cashflow

Infrastructure's contractual business nature helps provide investors with long-term visibility into cash flows with usually less exposure to price volatility and volume risk



Inflation Protection

Private Infrastructure cash flow are usually indexed to inflation. There are also often pass-through clauses for higher costs given the essential nature of their underlying business



Diversification

Private Infrastructure has a relatively low volatility relative to, and correlation with, traditional asset classes



Sustainability

The global shift towards net-zero suggests that investment into supporting infrastructure will increase as countries push to overhaul their existing carbon-based infrastructure

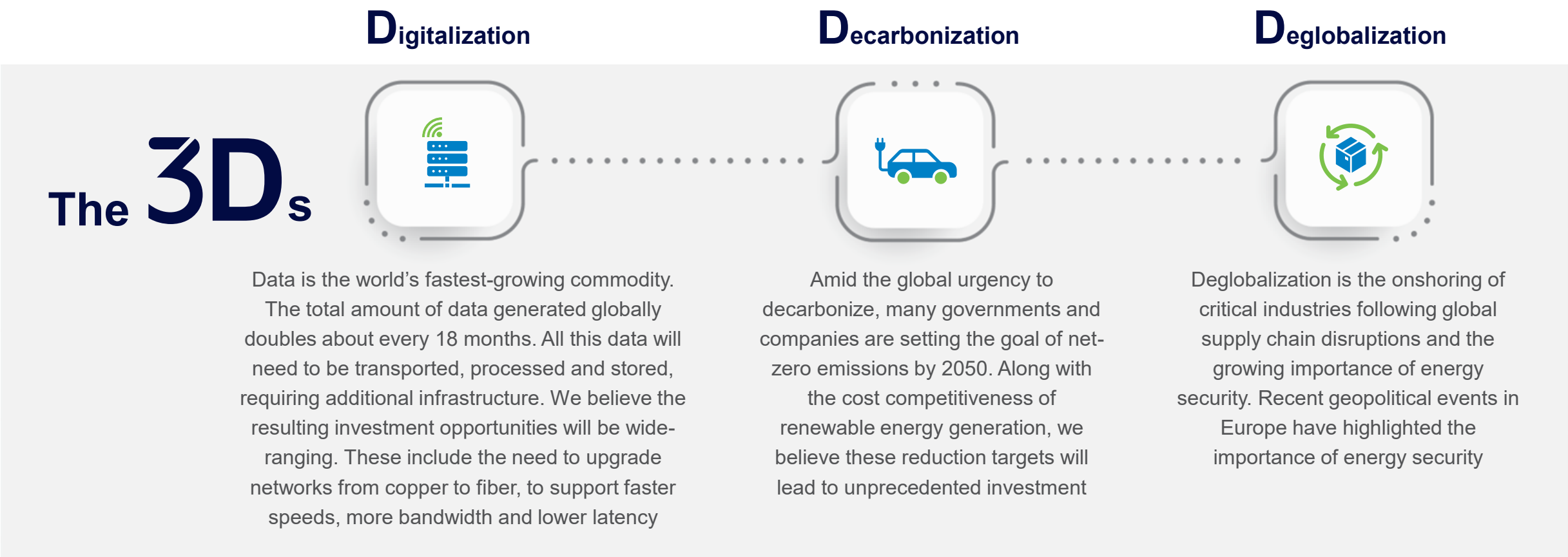
Source: Standard Chartered



Key Drivers of Infrastructure spending

The 3Ds of Infrastructure Opportunities

The infrastructure needs to enable long-term trends, including the investment to meet the staggering growth in data consumption, the shift away from traditional fossil fuels and the acceleration toward e-commerce, require a tremendous amount of capital

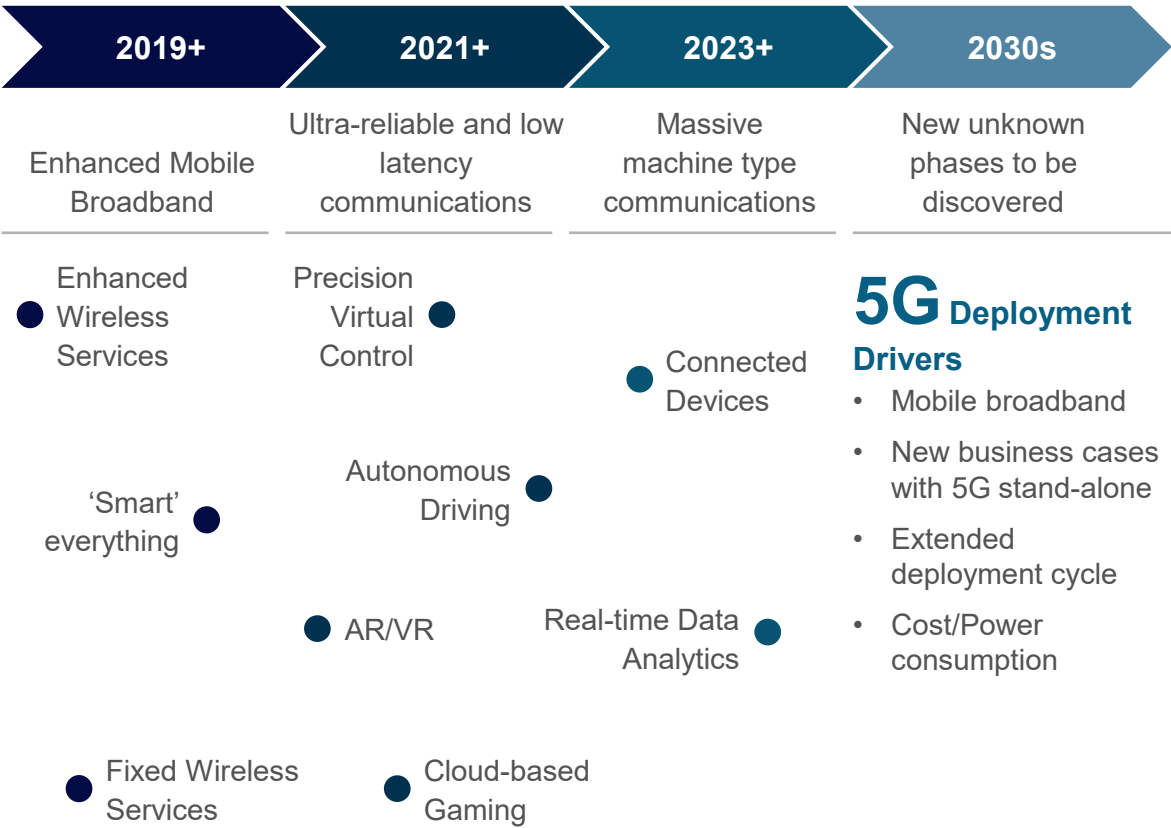


Source: Brookfield, Standard Chartered

Greater 5G adoption to drive spending on digital infrastructure

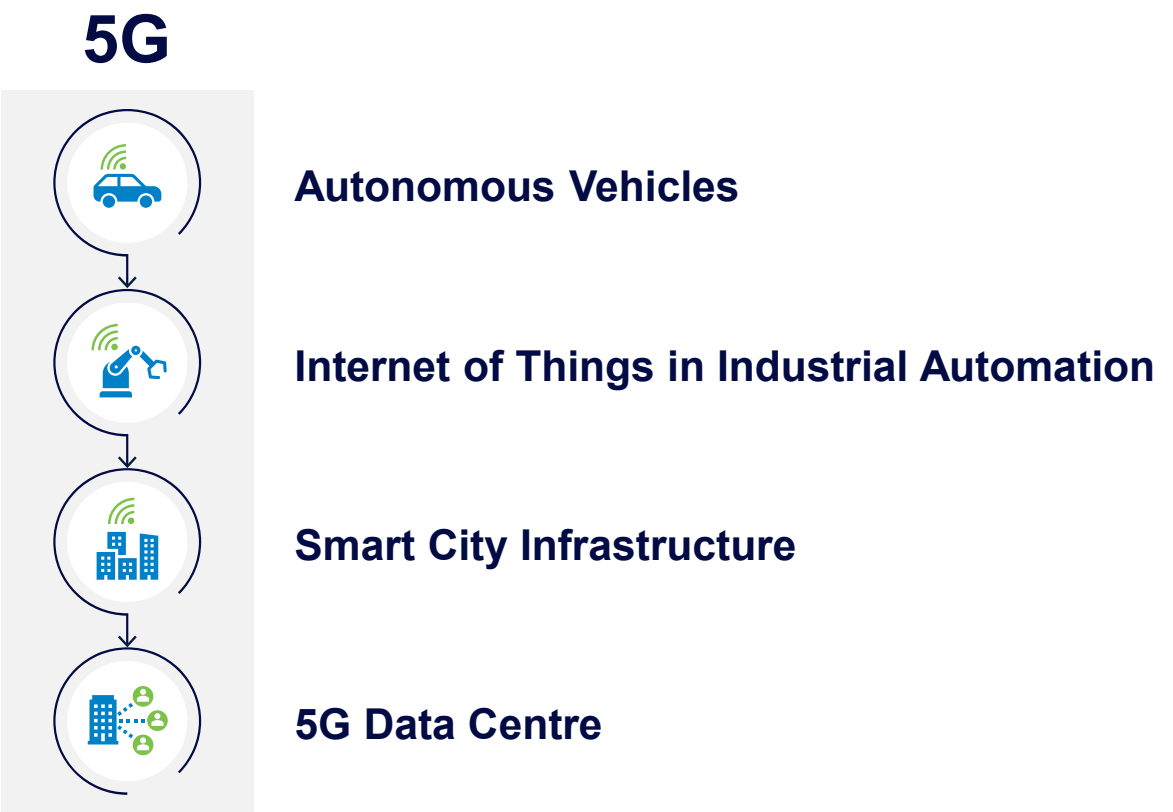
The construction of higher speed, lower latency 5G networks is expected to open a host of applications that previously were not viable. These applications are likely to drive Digital Infrastructure spending

5G Development Stages



Source: NXP Semiconductor, Standard Chartered

Infrastructure needs to support the Application of 5G

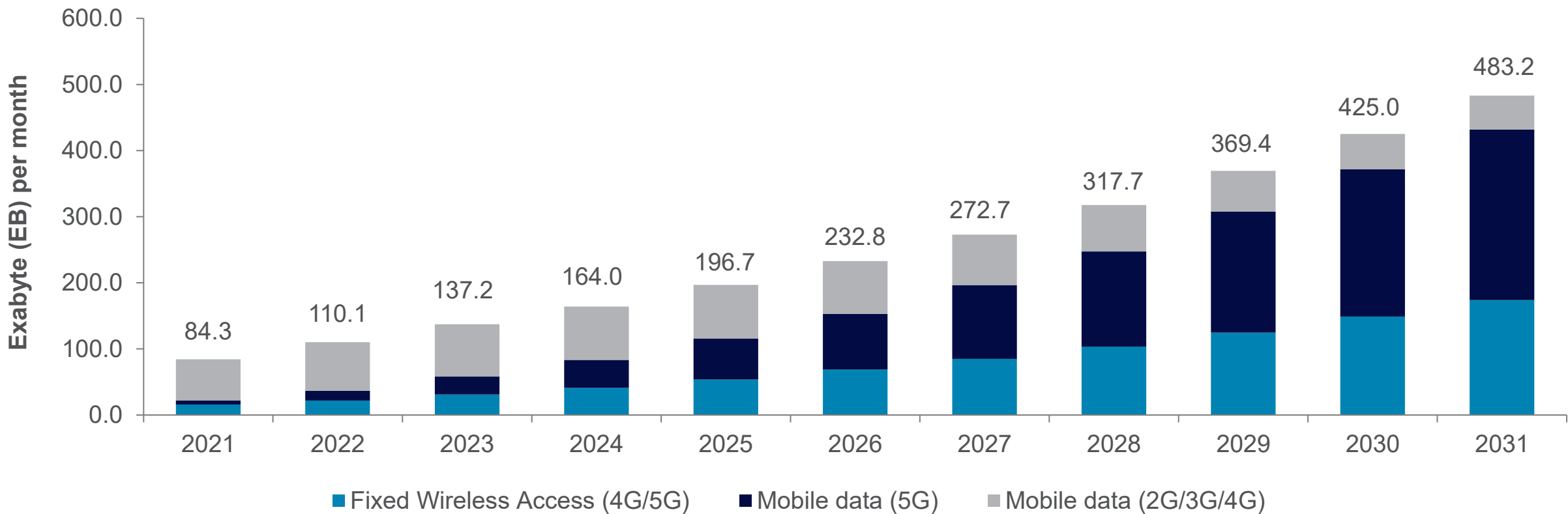


Source: Digi, Standard Chartered

The future of Data Consumption

Mobile data network consumption is expected to grow steadily and the need for more data centers and faster network infrastructure will drive investment needs

Global mobile network data traffic and forecasts



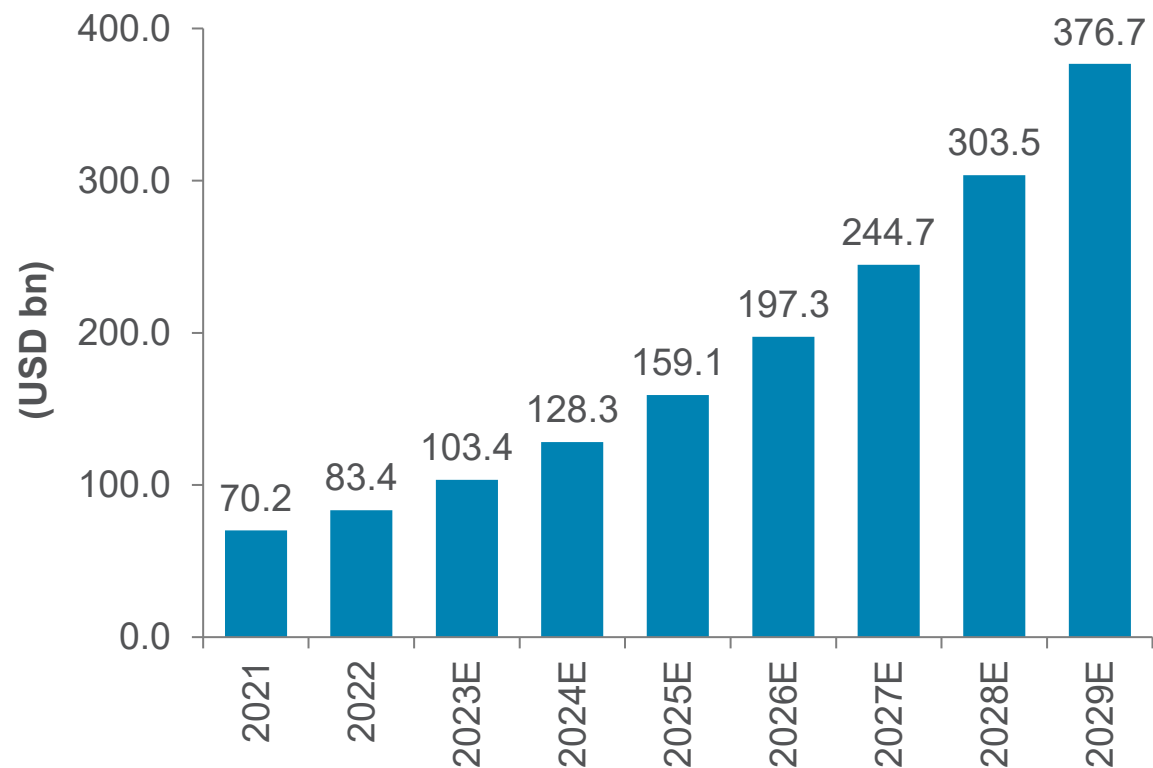
Source: Ericsson Mobility Report, Standard Chartered



The future of Digitalization

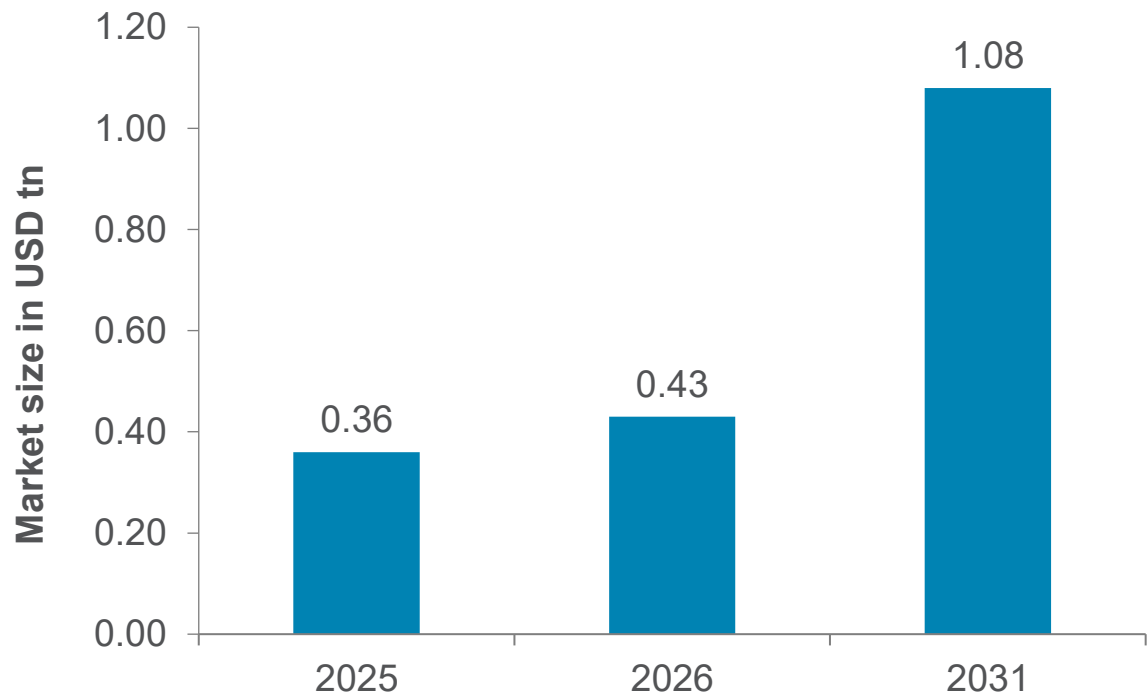
As the world marches towards digitalization, the growth in Digital Infrastructure spending is set to persist

Growth of Cloud Storage Market size



Source: Upwork, Synergy Research Group, Global X, Statista, Standard Chartered

Digital infrastructure markets are expected to grow at a CAGR of 20% from 2026 to 2031



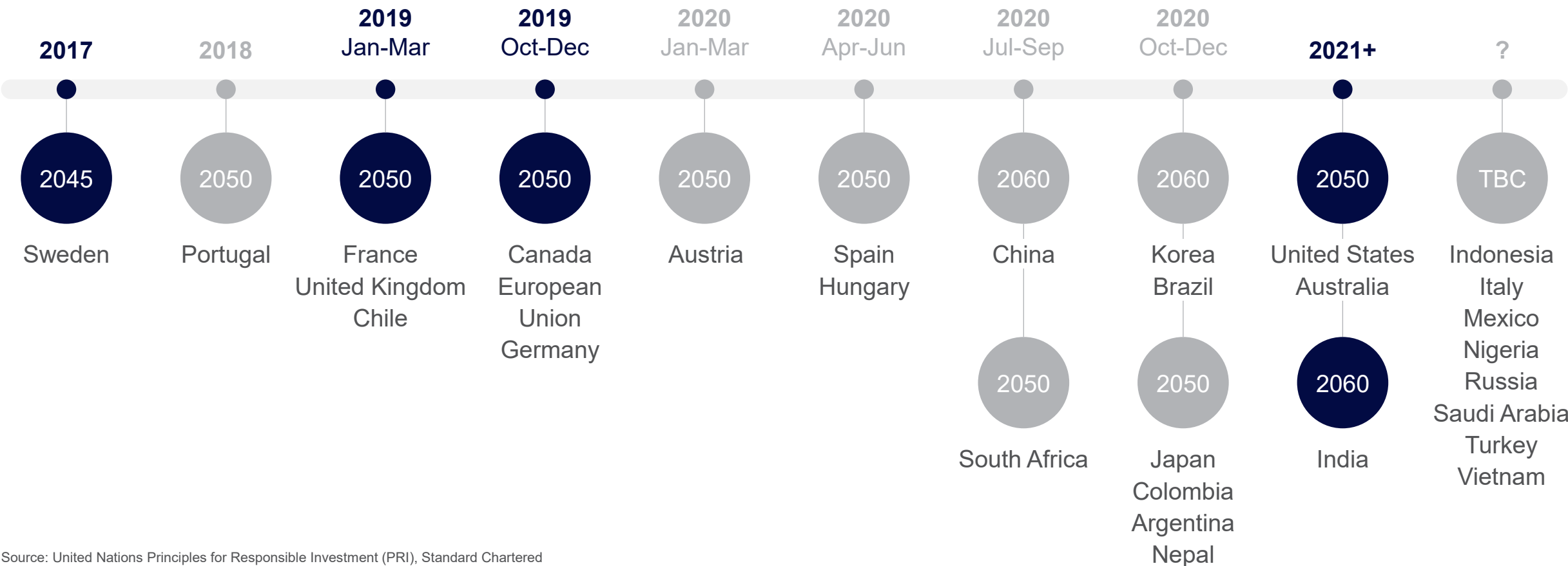
Source: Mordor Intelligence, Standard Chartered



Government pledges toward net zero carbon emissions

As the world look towards decarbonizing in the decades ahead, policies and investments are needed to overhaul existing infrastructure to achieve net-zero targets

Announcement timeline for net zero emissions targets



Source: United Nations Principles for Responsible Investment (PRI), Standard Chartered



Unprecedented investment in the build-out of renewable infrastructure

Many reduction targets will require significant capital investment to overhaul the power generation sector, the world's largest source of carbon emissions

Expected investment needed to transition energy systems by 2050

USD 150 trillion

Total investment to transition energy system by 2050



Build out renewable
power generation



Global
Electrification



Enhance electrical
infrastructure



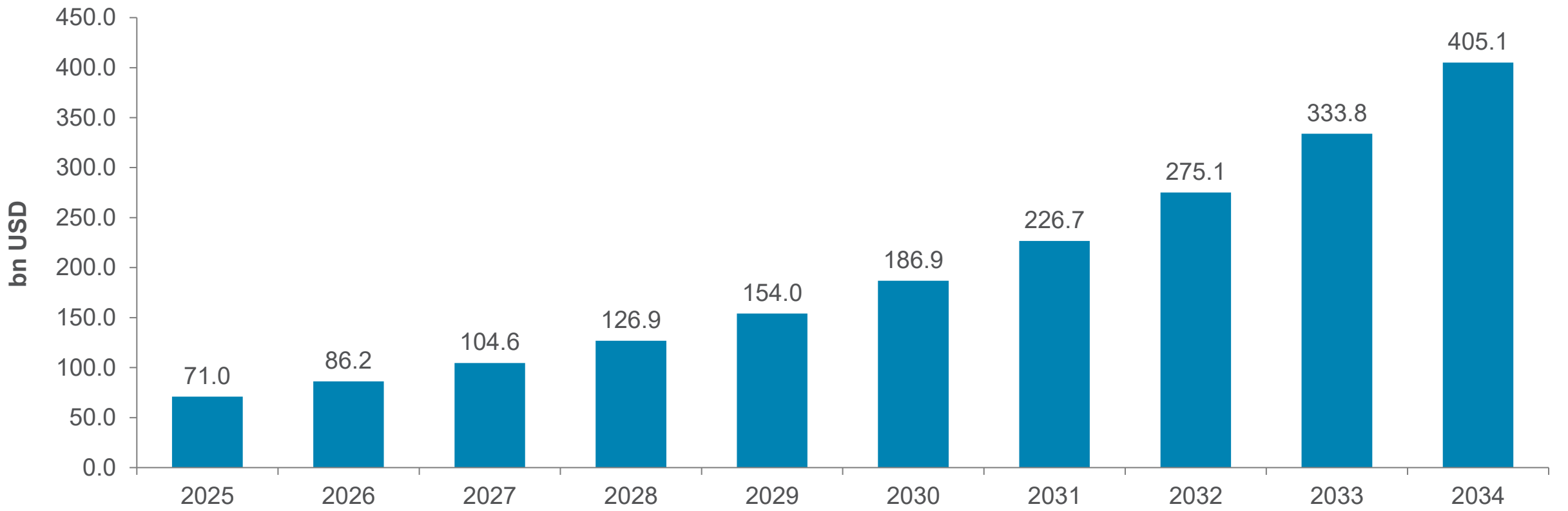
Repurpose existing
infrastructure assets

Source: Brookfield, Standard Chartered

The renewable infrastructure rush

Rising environmental awareness and government pressure for carbon-neutral construction methods are driving market growth for sustainable infrastructure

Sustainable Infrastructure Market Size



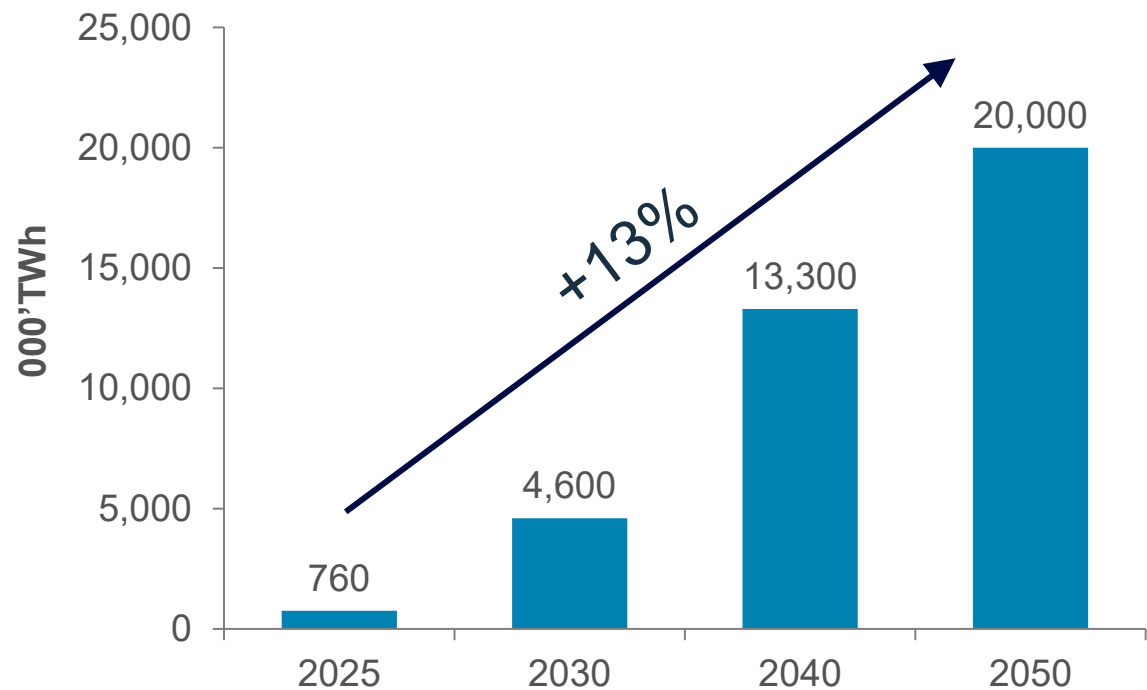
Source: Precedence Research, Standard Chartered



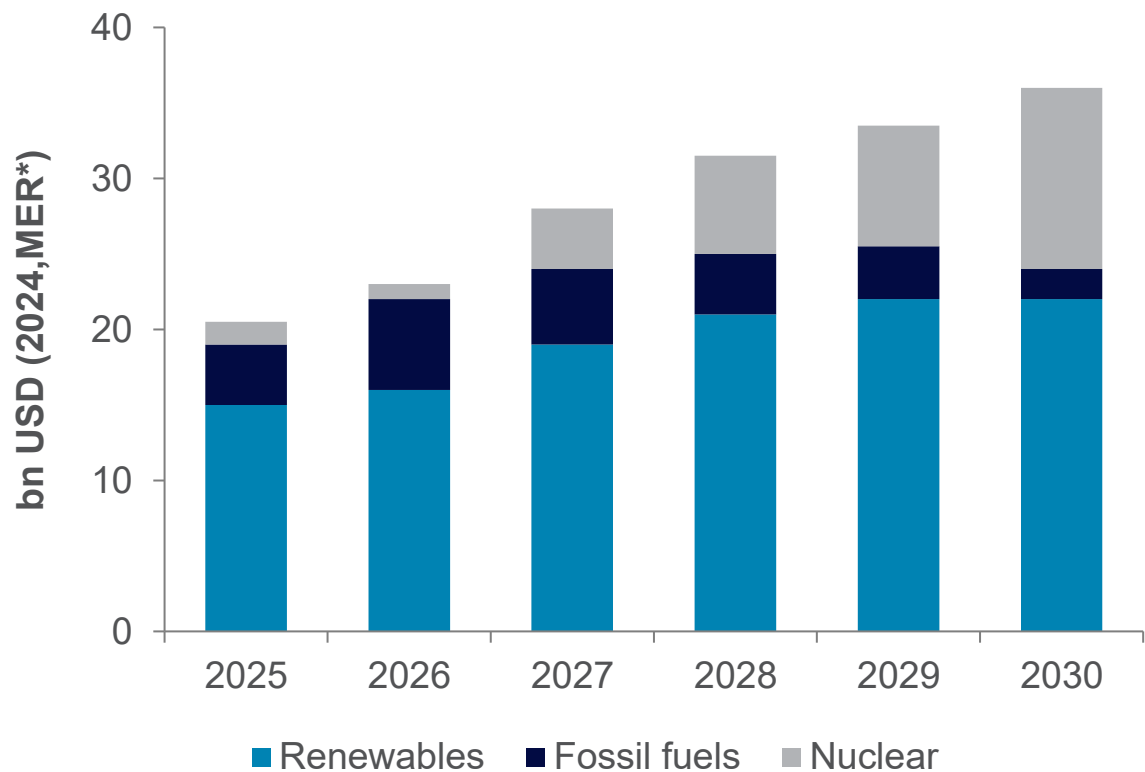
Acceleration of demand for global electricity

Data centres are driving demand for energy, and renewable energy infrastructure is meeting it

Global Electricity Demand Growth, with data centers projected 17% annual growth between 2025 to 2050



New power generation investment for data centers



Source: Brookfield, Standard Chartered

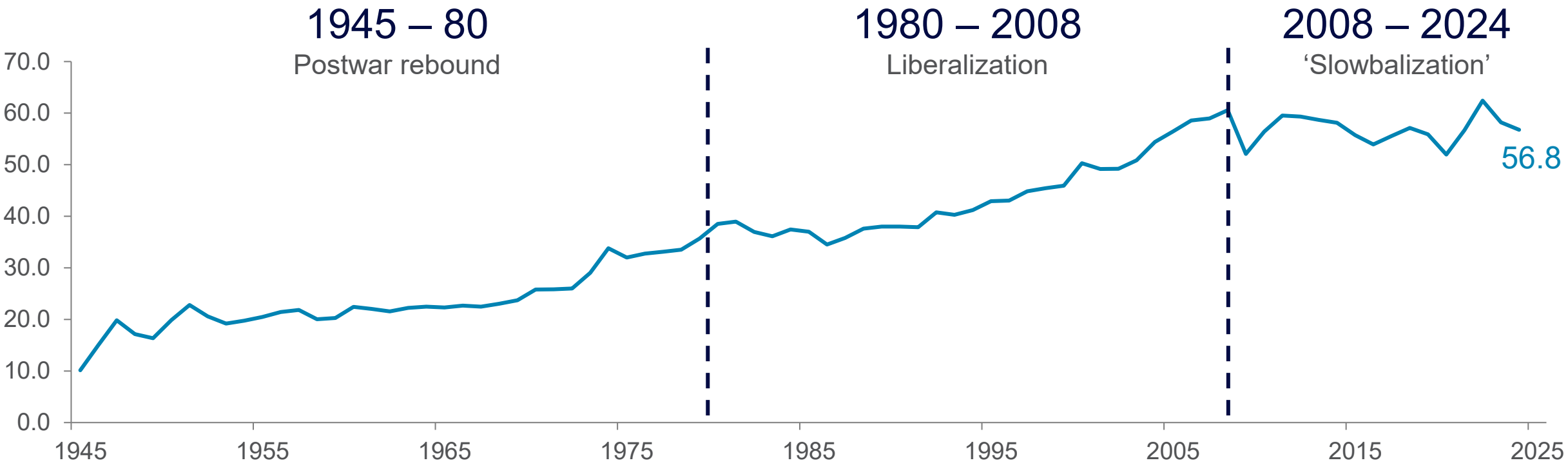
Source: International Energy Agency (IEA) World Investment Energy 2025 Report, Standard Chartered
* Market Exchange Rate



Globalization is in retreat for the first time since World War II

The pandemic has exposed the need for countries to onshore critical industries like energy security, supply chains and technology infrastructure

Global economy trade openness index

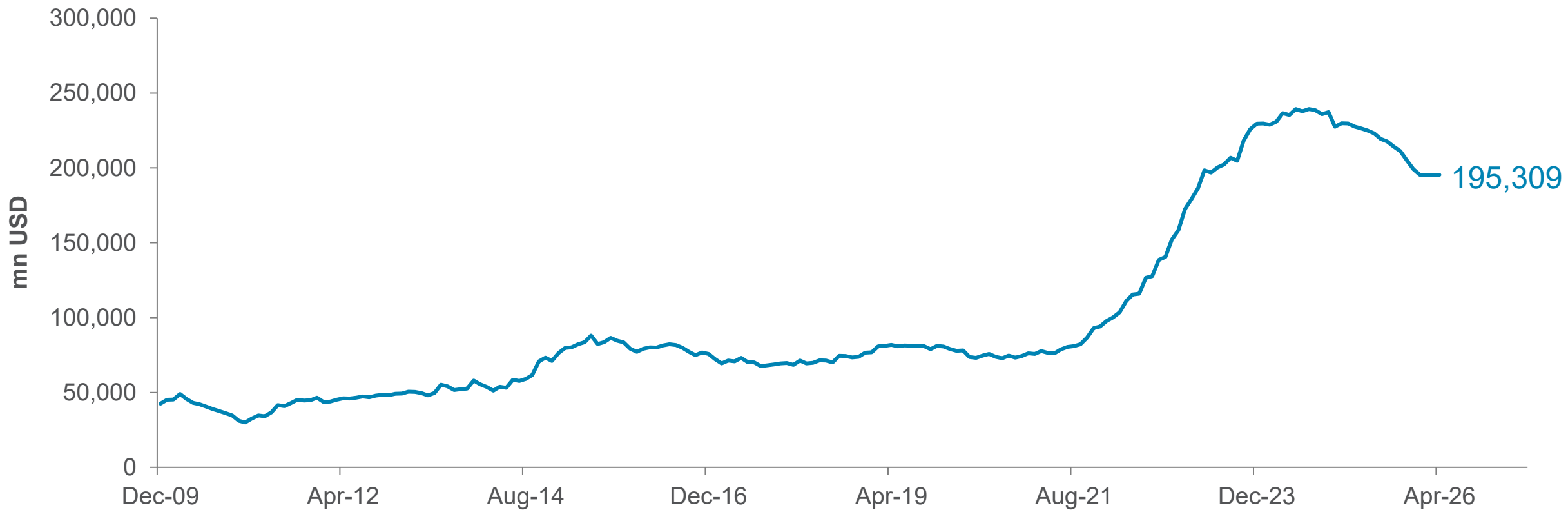


Source: Our World in Data, Peterson Institute for International Economics (PIIE), Standard Chartered
Note: National statistical organizations and central banks, OECD national accounts, and World Bank staff estimates (2026)Feenstra et al. - Penn World Table (2025)Klasing, Mariko J. and Milionis, P. (2014). "Quantifying the Evolution of World Trade, 1870-1949". Journal of International Economics 92(1), pp. 185-197Estevadeordal, A., Frantz, B., & Taylor, A. (2003). The Rise and Fall of World Trade, 1870-1939. The Quarterly Journal of Economics, 118(2), 359-407. Retrieved from <http://www.jstor.org/stable/25053910> – with major processing by Our World in Data

Rising geopolitical tension a threat to globalization

Geopolitical tensions is one of the drivers expected to drive reshoring efforts

Total US Private Construction Spending on Manufacturing



Source: Bloomberg, Standard Chartered



Takeaways: Decarbonization, Deglobalization and Digitalization to support spending on infrastructure assets

Key Drivers



Digitalization

Governments have earmarked digital infrastructure as a key focus area in their stimulus plans. Digital infrastructure is the foundation on which digital economic activities, technological applications and significant productivity gains can be built on.

Decarbonization

A convergence of interest in climate change from governments, corporates and individuals is accelerating the transition to a 'net zero' economy, of which the power and transport sector have a key role to play.

Deglobalization

Recent geopolitical events have highlighted the importance of onshoring critical industries. The pandemic has also revealed the inefficiencies and un-reliabilities in the existing supply chain infrastructure.



Key Risks

Key Risks

Risks	Summary
01 Political and Regulatory Risk	• Infrastructure assets are essential services that ensures the functioning of society and the modern economy; hence governments tend to maintain certain control or regulation over them • The level of exposure to political developments as well as the tightness of regulation differs from one country or sector to another and can have a significant impact on investment returns
02 Construction Risk	• In greenfield projects, construction risk are involved. Such risk includes project overruns and delays. These risks can be common for complex projects such as bridges or energy plants. Any project overruns or delays will usually cause delayed revenues and reduced returns
03 Operational Risk	• When an infrastructure asset is operating, operational risks will be involved, including performance or maintenance risks. Performance risk can include falling short of volume or price objectives or increased costs of doing business. Maintenance risk includes unplanned maintenance costs that can reduce operating cash flow levels. Exposure to operational risk varies by asset type • Regulated assets like electricity grids are typically more protected against certain risks through regulatory formulas or cost pass through to the customer, while unregulated assets are usually more exposed
04 Financial Risk	• Depending on the level of leverage and the debt structure, infrastructure assets can be exposed to interest rate volatility or refinancing risk. If not hedged, interest costs may increase, reducing operating cash flow levels

Source: Deutsche Asset Management, Standard Chartered



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